

Estimates for Weyl exponential sums

Yule Lee (yulelee07@163.com)

May, 2025

Let $f(x)$ and $g(x)$ be real functions defined on the interval $[a, b]$ with sufficiently many derivatives ($b - a \geq 1$). This paper will discuss the upper bound estimation of exponential sums

$$\sum_{a < n \leq b} e(f(n)), \quad \sum_{a < n \leq b} g(n)e(f(n)).$$

Here $e(f(n)) = e^{2\pi i f(n)}$.

H. Weyl (1916) first provided non-trivial upper bound estimates for such exponential sums in his pioneering work on uniform distribution. After that, exponential sums appeared in important research topics of analytic number theory such as Riemann function theory, prime number distribution, Waring's problem, lattice point problems, and Diophantine approximation, and estimating such exponential sums became the key to studying these problems. Van der Corput and I. M. Vinogradov proposed two new methods for estimating Weyl sums, which are superior to Weyl's method (Since then, combining the ideas of these methods, Bombieri and Iwaniec proposed a new method for estimating individual exponential sums by estimating the mean value of exponential sums, and successfully applied it to estimate the order of $\zeta(\frac{1}{2} + it)$). In practical applications, $g(x)$ is a monotone function, and $a < b \leq 2a$, hence by partial summation, we only need to discuss the first exponential sum.

I. van der Corput Method

The core idea of the van der Corput method is to transform the exponential sum into a sum of exponential integrals (Theorem 1.5), i.e., into the exponential integral

$$\int_a^b e^{i(f(x) - vx)} dx = \int_a^b e^{iF(x)} dx,$$

summed over v . He found that if $|F'(x)| \geq r > 0$ (note that $F'(x) = f'(x)$, r and v are independent), we can obtain a good upper bound estimate for the exponential integral (Lemma 2.1), which is precisely why his method is superior to Weyl's method. Thus, when $f''(x)$ satisfies certain conditions, an upper bound estimate for the exponential sum is obtained (Theorem 2.3). When $f(x)$ does not satisfy the required conditions, the exponential sum can be transformed into other exponential sums through two approaches (Theorems 3.1 and 5.2). By repeatedly applying these two approaches (which can be combined), until the second derivative of the function in the final set of exponential sums, i.e. the new $f(x)$, satisfies the conditions of Theorem 2.3, we can estimate their upper bounds. This yields the upper bound estimate for the original exponential sum. These are the contents discussed in Sections 4 and 6.

1 Fundamental Relations

Lemma 1.1. *Suppose $f'(x)$ is continuous on the closed interval $[a, b]$. Then we have*

$$\sum_{a < n \leq b} e(f(n)) = \int_a^b e(f(x)) dx - b_1(x)e(f(x)) \Big|_a^b - \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f'(x) \{e(f(x) + \nu x) - e(f(x) - \nu x)\} dx, \quad (1)$$

where $b_1(x) = x - [x] - \frac{1}{2}$.

Proof. By the Euler-Maclaurin summation formula, we obtain

$$\sum_{a < n \leq b} e(f(n)) = \int_a^b e(f(x)) dx + \int_a^b b_1(x) 2\pi i f'(x) e(f(x)) dx - b_1(x)e(f(x)) \Big|_a^b.$$

Clearly, the function $b_1(x)$ is an odd periodic function with period 1, and $\int_0^1 b_1(x) du = 0$. then the Fourier series of

$b_1(x)$ can be written as $-\frac{1}{2\pi i} \sum_{\nu=1}^{\infty} \frac{1}{\nu} (e(\nu x) - e(-\nu x))$, which converges boundedly for all x , and converges uniformly

$b_1(x)$ on any closed interval not containing integers. Therefore, substituting the Fourier series of $b_1(x)$ into the second integral on the right-hand side of the previous formula and integrating term-by-term, we obtain (1). $\square$

Completely analogously, we have

Lemma 1.2. Suppose both $f(x)$ and $g(x)$ are continuous on the closed interval $[a, b]$. Then we have

$$\begin{aligned} \sum_{a < n \leq b} g(n)e(f(n)) &= \int_a^b g(x)e(f(x))dx - b_1(x)g(x)e(f(x)) \Big|_a^b \\ &\quad - \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b \left(g(x)f'(x) + \frac{g'(x)}{2\pi i} \right) \{e(f(x) + vx) - e(f(x) - vx)\} dx. \end{aligned} \quad (2)$$

In order to estimate the integrals in (1) and (2) which can be treated as error terms, we need

Lemma 1.3. Suppose on the interval $[a, b]$, the function $\frac{G(x)}{F'(x)}$ is monotone, and satisfies $\frac{F'(x)}{G(x)} \geq m > 0$ (or $-\frac{F'(x)}{G(x)} \geq m > 0$). Then we have $\left| \int_a^b G(x)e^{iF(x)} dx \right| \leq \frac{4}{m}$.

Proof. First assume that $\frac{G(x)}{F'(x)}$ is positive and decreasing. By the second mean value theorem for integrals, we obtain:

$$\int_a^b G(x) \cos F(x) dx = \int_a^b \frac{G(x)}{F'(x)} F'(x) \cos F(x) dx = \frac{G(a)}{F'(a)} \int_a^c F'(x) \cos F(x) dx = \frac{G(a)}{F'(a)} (\cos F(c) - \cos F(a)),$$

where $a \leq c \leq b$. Thus, $\left| \int_a^b G(x) \cos F(x) dx \right| \leq \frac{2}{m}$. Similarly, we can prove $\left| \int_a^b G(x) \sin F(x) dx \right| \leq \frac{2}{m}$. Combining the above two formulas yields the desired conclusion. Other cases can be proved similarly. $\square$

Similar to Lemma 1.3, it is easy to prove the following lemma.

Lemma 1.4. Suppose on the interval $[a, b]$, $H(x)$ is monotone and $|H(x)| \leq M$. Then, under the conditions of Lemma 1.3, we have

$$\left| \int_a^b H(x)G(x)e^{iF(x)} dx \right| \leq \frac{8M}{m}. \quad (3)$$

Proof. First assume that $\frac{G(x)}{F'(x)}$ is a positive and decreasing function. By the second mean value theorem for integrals, we obtain

$$\int_a^b H(x)G(x) \cos F(x) dx = H(a) \int_a^c G(x) \cos F(x) dx + H(b) \int_c^b G(x) \cos F(x) dx.$$

From this we obtain $\left| \int_a^b H(x)G(x) \cos F(x) dx \right| \leq \frac{4M}{m}$. Similarly, we can prove $\left| \int_a^b H(x)G(x) \sin F(x) dx \right| \leq \frac{4M}{m}$. Combining the above two formulas yields the desired conclusion. Other cases can be proved similarly. $\square$

Next, we prove the fundamental relations.

Theorem 1.5. Suppose $f(x)$ is monotone and continuous on the interval $[a, b]$, with $f'(a) = \beta$ and $f'(b) = \alpha$. Let η be a real number such that $0 < \eta < 1$. Then, if $f(x)$ is decreasing we have

$$\sum_{a < n \leq b} e(f(n)) = \sum_{\alpha - \eta < \nu < \beta + \eta} \int_a^b e(f(x) - vx) dx + O\left(\frac{1}{\eta} + \log(\beta - \alpha + 2)\right). \quad (4)$$

When $f(x)$ is increasing, we have

$$\sum_{a < n \leq b} e(f(n)) = \sum_{\beta - \eta < \nu < \alpha + \eta} \int_a^b e(f(x) - vx) dx + O\left(\frac{1}{\eta} + \log(\alpha - \beta + 2)\right). \quad (5)$$

Proof. First prove (4). For a given η , there must be a unique integer k such that $\eta \leq \alpha + k < \eta + 1$. Let $f_1(x) = f(x) + kx$, $\alpha_1 = \alpha + k$, $\beta_1 = \beta + k$. Thus, proving (4) is equivalent to proving

$$\sum_{a < n \leq b} e(f_1(n)) = \sum_{1 \leq \nu < \beta_1 + \eta} \int_a^b e(f_1(x) - vx) dx + O\left(\frac{1}{\eta} + \log(\beta_1 - \alpha_1 + 2)\right). \quad (6)$$

From Lemma 1.1, we know

$$\sum_{a < n \leq b} e(f_1(n)) = \int_a^b e(f_1(x))dx - \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f_1'(x) \{e(f_1(x) + vx) - e(f_1(x) - vx)\}dx + O(1). \quad (7)$$

Since the function $f_1'(x)$ is also decreasing, and

$$\eta \leq f_1'(b) = \alpha_1 < \eta + 1, \quad \alpha_1 \leq f_1'(x) \leq \beta_1 = f_1'(a). \quad (8)$$

Thus, when $\nu \geq 1$, the function $\frac{f_1'(x)}{f_1'(x) + \nu} = 1 - \frac{\nu}{f_1'(x) + \nu}$ is positive and decreasing on the interval $[a, b]$, and

$$0 < \frac{\alpha_1}{\alpha_1 + \nu} \leq \frac{f_1'(x)}{f_1'(x) + \nu} \leq \frac{\beta_1}{\beta_1 + \nu}, \quad a \leq x \leq b.$$

Thus, by Lemma 1.3 (taking $G(x) = f_1'(x)$ and $F(x) = 2\pi(f_1(x) + vx)$), we obtain

$$\left| \int_a^b f_1'(x) e(f_1(x) + vx) dx \right| \leq \frac{4\beta_1}{2\pi(\beta_1 + \nu)}, \quad \nu \geq 1.$$

Furthermore, we have

$$\left| \sum_{\nu=1}^{\infty} \frac{1}{\nu} \int_a^b f_1'(x) e(f_1(x) + vx) dx \right| \leq \frac{4}{2\pi} \sum_{\nu=1}^{\infty} \frac{\beta_1}{\nu(\nu + \beta_1)} \leq \frac{4\beta_1}{2\pi(1 + \beta_1)} + \frac{4}{2\pi} \log(1 + \beta_1) \ll \log(\beta_1 - \alpha_1 + 2). \quad (9)$$

When $\nu \geq \beta_1 + \eta$, since $f_1'(x)$ is decreasing, the function $\frac{f_1'(x)}{f_1'(x) - \nu} = 1 + \frac{\nu}{f_1'(x) - \nu}$ is negative and increasing, and

$$\frac{\beta_1}{\beta_1 - \nu} \leq \frac{f_1'(x)}{f_1'(x) - \nu} \leq \frac{\alpha_1}{\alpha_1 - \nu} < 0.$$

Therefore, from Lemma 1.3 (taking $G(x) = f_1'(x)$, $F(x) = 2\pi(f_1(x) - vx)$), we get

$$\left| \int_a^b f_1'(x) e(f_1(x) - vx) dx \right| \leq \frac{4\beta_1}{2\pi(\nu - \beta_1)}, \quad \nu \geq \beta_1 + \eta.$$

Furthermore, we have

$$\begin{aligned} \left| \sum_{\nu \geq \beta_1 + \eta} \frac{1}{\nu} \int_a^b f_1'(x) e(f_1(x) - vx) dx \right| &\leq \frac{4}{2\pi} \sum_{\nu \geq \beta_1 + \eta} \frac{\beta_1}{\nu(\nu - \beta_1)} \\ &\leq \frac{4}{2\pi} \frac{\beta_1}{\eta(\beta_1 + \eta)} + \frac{4}{2\pi} \log \frac{\beta_1 + \eta}{\eta} \ll \frac{1}{\eta} + \log(\beta_1 - \alpha_1 + 2). \end{aligned} \quad (10)$$

Moreover, from Lemma 1.3 (taking $G(x) \equiv 1$, $F(x) = 2\pi f_1(x)$), we get $\int_a^b e(f_1(x))dx \ll \frac{1}{\eta}$. From (7), (9), and (10), it is proved that

$$\sum_{a < n \leq b} e(f_1(n)) = \sum_{1 \leq \nu < \beta_1 + \eta} \frac{1}{\nu} \int_a^b f_1'(x) e(f_1(x) - vx) dx + O\left(\frac{1}{\eta} + \log(\beta_1 - \alpha_1 + 2)\right).$$

From this and

$$\frac{1}{\nu} \int_a^b f_1'(x) e(f_1(x) - vx) dx = \int_a^b e(f_1(x) - vx) dx + O\left(\frac{1}{\nu}\right),$$

we obtain (6), which in turn proves (4).

When $f(x)$ is increasing, we may consider the function $f_2(x) = -f(x)$. Then (4) holds for $f_2(x)$. Taking the complex conjugate and substituting v by $-v$, we obtain (5). $\square$

We can also prove the following theorem by the same method.

Theorem 1.6. Suppose $g(x)$ and $g'(x)$ are both monotone and continuous on the interval $[a, b]$, and $|g(x)| \leq h_0$, $|g'(x)| \leq h_1$. Then, under the conditions of Theorem 1.5, when $f(x)$ is decreasing, we have

$$\sum_{a < n \leq b} g(n)e(f(n)) = \sum_{\alpha - \eta < \nu < \beta + \eta} \int_a^b g(x)e(f(x) - vx)dx + O\left(\frac{h_0 + h_1}{\eta} + h_0 \log(\beta - \alpha + 2)\right); \quad (11)$$

when $f(x)$ is increasing, we have

$$\sum_{a < n \leq b} g(n)e(f(n)) = \sum_{\beta - \eta < \nu < \alpha + \eta} \int_a^b g(x)e(f(x) - vx)dx + O\left(\frac{h_0 + h_1}{\eta} + h_0 \log(\alpha - \beta + 2)\right). \quad (12)$$

It should be noted that in the proof wherever Lemma 1.1 and Lemma 1.3 are used in the proof of Theorem 1.5, they should be replaced by Lemma 1.2 and Lemma 1.4, respectively.

As useful special cases, we have

Theorem 1.7. Suppose on the interval $[a, b]$, we have $|f'(x)| \leq \theta < 1$. Then,

(a) Under the conditions of Theorem 1.5, we have

$$\sum_{a < n \leq b} e(f(n)) = \int_a^b e(f(x))dx + O\left(\frac{1}{1 - \theta}\right); \quad (13)$$

(b) Under the conditions of Theorem 1.6, we have

$$\sum_{a < n \leq b} g(n)e(f(n)) = \int_a^b g(x)e(f(x))dx + O\left(\frac{h_0 + h_1}{1 - \theta}\right). \quad (14)$$

Proof. We prove (13) here. Equation (14) can be proven similarly. Take $\eta = \frac{1}{2}(1 - \theta)$.

When $f(x)$ is decreasing, since

$$\alpha - \eta \geq -\theta - \frac{1}{2}(1 - \theta) > -1, \quad \beta + \eta \leq \theta + \frac{1}{2}(1 - \theta) < 1,$$

the sum on the right-hand side of (4) contains at most one term corresponding to $\nu = 0$. If this term appears, then (13) is already proven; if this term does not appear, then either of the following two cases must hold:

- (1) $\alpha - \eta > 0$, which means $f'(x) \geq \alpha > \eta = \frac{1}{2}(1 - \theta)$;
- (2) $\beta + \eta < 0$, which means $f'(x) \leq \beta < -\eta = -\frac{1}{2}(1 - \theta)$.

In either case, from Lemma 1.3 (taking $G(x) \equiv 1$), we get $\int_a^b e(f(x))dx \ll \frac{1}{1 - \theta}$. Therefore, from (4), (13) also holds. The same proof applies when $f'(x)$ is increasing (or by considering $-f(x)$ and then taking the complex conjugate). $\square$

2 Basic Estimates

From Lemma 1.3 (taking $G(x) \equiv 1$), we know that when $F'(x)$ is monotone and $|F'(x)| \geq m > 0$ ($a \leq x \leq b$), the exponential integral has the estimate

$$\int_a^b e^{iF(x)}dx \ll \min\left(b - a, \frac{1}{m}\right). \quad (15)$$

This also indicates that the main part of a general exponential integral (C) lies in the small interval centered at the point x where $F'(x) = 0$. The exponential integrals appearing on the right-hand sides of (4) and (5) are precisely of this type. To estimate such exponential integrals, van der Corput proved the following fundamental lemma, which is the foundation of his method for estimating Weyl sums.

Lemma 2.1. Suppose $F''(x)$ is continuous on the interval $[a, b]$, and $|F''(x)| \geq r > 0$. Then we have

$$\int_a^b e^{iF(x)}dx \ll \min(b - a, r^{-\frac{1}{2}}). \quad (16)$$

Proof. Without loss of generality, assume $F''(x) \geq r > 0$; otherwise, consider $-F(x)$. Then $F'(x)$ is strictly increasing, and there is at most one point c in the interval $[a, b]$ such that $F'(c) = 0$. We discuss the two cases separately:

(I) The case where there is a zero point c . Let δ be a constant to be determined, and we split the integral as follows: $\int_a^b e^{iF(x)} dx = \int_a^{c-\delta} + \int_{c-\delta}^{c+\delta} + \int_{c+\delta}^b$. Since

$$F'(x) = - \int_x^c F''(u) du \leq -r(c-x) \leq -r\delta, \quad a \leq x \leq c-\delta,$$

$$F'(x) = \int_c^x F''(u) du \geq r\delta > 0, \quad c+\delta \leq x \leq b,$$

by Lemma 1.3, we have

$$\int_a^{c-\delta} e^{iF(x)} dx \ll \frac{1}{r\delta}, \quad \int_{c+\delta}^b e^{iF(x)} dx \ll \frac{1}{r\delta}.$$

Hence we have $\int_a^b e^{iF(x)} dx \ll \frac{1}{r\delta} + \delta$. Taking $\delta = r^{-\frac{1}{2}}$, we immediately obtain (16) (Note: when $c - r^{-\frac{1}{2}} < a$ or $c + r^{-\frac{1}{2}} > b$, the result obviously still holds).

(II) The case where there is no zero point c . Let δ be a constant to be determined. When $F'(x) > 0$, we have

$$F'(x) = F'(a) + \int_a^x F''(u) du > r\delta > 0, \quad a + \delta \leq x \leq b.$$

Therefore, from Lemma 1.3, we get

$$\int_a^b e^{iF(x)} dx = \int_a^{a+\delta} e^{iF(x)} dx + \int_{a+\delta}^b e^{iF(x)} dx \ll \delta + \frac{1}{r\delta}.$$

When $F'(x) < 0$, we have

$$F'(x) = F'(b) - \int_x^b F''(u) du < -r\delta < 0, \quad a \leq x \leq b - \delta.$$

Thus,

$$\int_a^b e^{iF(x)} dx = \int_a^{b-\delta} e^{iF(x)} dx + \int_{b-\delta}^b e^{iF(x)} dx \ll \frac{1}{r\delta} + \delta.$$

In either case, taking $\delta = r^{-\frac{1}{2}}$, we immediately obtain (16). (Note: when $a + r^{-\frac{1}{2}} > b$ or $b - r^{-\frac{1}{2}} < a$, (16) obviously still holds).

Similarly, using Lemma 1.4 we can easily prove the following lemma.

Lemma 2.2. Suppose $H(x)$ is monotone on the interval $[a, b]$ and $|H(x)| \leq M$, then under the conditions of Lemma 2.1 we have

$$\int_a^b H(x) e^{iF(x)} dx \ll \min \left(M(b-a), Mr^{-\frac{1}{2}} \right). \quad (17)$$

From Theorems 1.5 and Lemma 2.1, we can immediately obtain the basic estimate for the Weyl sum.

Theorem 2.3. Suppose $b - a \geq 1$. On the interval $[a, b]$, $f''(x)$ is continuous and satisfies $0 < \lambda_2 \leq |f''(x)| \leq h\lambda_2$ then we have

$$\sum_{a < n \leq b} e(f(n)) \ll h(b-a)\lambda_2^{\frac{1}{2}} + \lambda_2^{-\frac{1}{2}}, \quad (18)$$

where the constant implied by $\ll$ is an absolute constant.

Proof. When $\lambda_2 \geq 1$, estimate (18) is trivial. Thus, we may assume $\lambda_2 < 1$. Clearly, $f'(x)$ satisfies the conditions of Theorem 1.5. In Theorem 1.5, taking $\eta = \frac{1}{2}$, since $2\pi(f(x) - vx) = F(x)$ satisfies the conditions of Lemma 2.1, using Lemma 1 to estimate the right-hand integral of (4) we obtain

$$\sum_{a < n \leq b} e(f(n)) \ll (|\alpha - \beta| + 1)\lambda_2^{-\frac{1}{2}} + \log(|\alpha - \beta| + 2) \ll (|\alpha - \beta| + 1)\lambda_2^{-\frac{1}{2}}.$$

From this and

$$|\alpha - \beta| = |f'(b) - f'(a)| = \left| \int_a^b f''(x) dx \right| \leq h\lambda_2(b-a),$$

we immediately obtain (18). □

3 Basic Inequalities

Theorem 3.1. *Let l be a positive integer, $l \leq b - a$. Then we have*

$$\left| \sum_{a < n \leq b} e(f(n)) \right| \leq 2 \frac{b-a}{\sqrt{l}} + 2 \sqrt{\frac{b-a}{l}} \cdot \left\{ \sum_{r=1}^{l-1} \left| \sum_{a < n \leq b-r} e(f(n+r) - f(n)) \right| \right\}^{\frac{1}{2}}. \quad (19)$$

Proof. For the sake of simplicity, we define $e(f(n)) = 0$, when $n \leq a$ or $n > b$. Under this convention, $e(-f(n))$ denotes the complex conjugate of $e(f(n))$, not its reciprocal. Thus, we have

$$\begin{aligned} l \sum_{a < n \leq b} e(f(n)) &= l \sum_{n=-\infty}^{\infty} e(f(n)) = \sum_{m=1}^l \sum_{n=-\infty}^{\infty} e(f(m+n)) \\ &= \sum_{m=1}^l \sum_{q-m < n \leq b-m} e(f(m+n)) = \sum_{a-l < n \leq b-1} \sum_{m=1}^l e(f(m+n)). \end{aligned} \quad (20)$$

Applying the Cauchy-Schwarz inequality, we obtain

$$l^2 \left| \sum_{a < n \leq b} e(f(n)) \right|^2 \leq (b-a+l) \left\{ \sum_{a-l < n \leq b-1} \left| \sum_{m=1}^l e(f(m+n)) \right|^2 \right\}. \quad (21)$$

Since

$$\begin{aligned} \left| \sum_{m=1}^l e(f(m+n)) \right|^2 &= \sum_{m=1}^l \sum_{m'=1}^l e(f(m+n) - f(m'+n)) \\ &= l + 2 \operatorname{Re} \sum_{m=2}^l \sum_{m'=1}^{m-1} e(f(m+n) - f(m'+n)), \end{aligned}$$

then

$$l^2 \left| \sum_{a < n \leq b} e(f(n)) \right|^2 \leq l(b-a+l)^2 + 2(b-a+l) \operatorname{Re} \left\{ \sum_{a-l < n \leq b-1} \sum_{m=2}^l \sum_{m'=1}^{m-1} e(f(m+n) - f(m'+n)) \right\}.$$

$$\begin{aligned} &\sum_{a-l < n \leq b-1} \sum_{m=2}^l \sum_{m'=1}^{m-1} e(f(m+n) - f(m'+n)) \\ &= \sum_{m=2}^l \sum_{m'=1}^{m-1} \sum_{n=-\infty}^{\infty} e(f(m+n) - f(m'+n)) = \sum_{m=2}^l \sum_{m'=1}^{m-1} \sum_{n'=-\infty}^{\infty} e(f(m-m'+n') - f(n')) \\ &= \sum_{m=2}^l \sum_{r=1}^{m-1} \sum_{n'=-\infty}^{\infty} e(f(n'+r) - f(n')) = \sum_{r=1}^{l-1} \sum_{m=r+1}^l \sum_{n'=-\infty}^{\infty} e(f(n'+r) - f(n')) \\ &= \sum_{r=1}^{l-1} (l-r) \sum_{a < n \leq b-r} e(f(n+r) - f(n)). \end{aligned}$$

From the above two formulas, we derive

$$l \left| \sum_{a < n \leq b} e(f(n)) \right|^2 \leq (b-a+l) \left\{ b-a+l + 2 \operatorname{Re} \left(\sum_{r=1}^{l-1} \left(1 - \frac{r}{l}\right) \sum_{a < n \leq b-r} e(f(n+r) - f(n)) \right) \right\}. \quad (22)$$

This proves (19). $\square$

The significance of this inequality is that it transforms the estimation of the exponential sum $\sum_{a < n \leq b} e(f(n))$ into

the estimation of the exponential sum $\sum_{c < n \leq d} e(J(n))$, where $J(x) = f(x+r) - f(x)$. Here r is a parameter. From the

mean value theorem, the conditions imposed on the k -th derivative of $f(x)$ are equivalent to those imposed on the $(k-1)$ -th derivative of $J(x)$. Thus, by repeatedly applying this fundamental inequality, we can transform the problem

into estimating the exponential sum when the second derivative of $J(x)$ satisfies certain conditions. For this type of exponential sum, Theorem 2.3 already gives a direct estimate. This is the core idea of the van der Corput method for estimating Weyl sums.

This fundamental inequality is due to Weyl, and it is derived from (22). However, the right-hand side of (22) actually involves estimating a double exponential sum. This leads to the study of double and multiple exponential sums.

4 Weyl Sum Estimates

Theorem 4.1. *Let integer $k \geq 2$, $K = 2^{k-1}$. Assume $b - a \geq 1$. On the interval $[a, b]$, $f(x)$ has continuous derivatives of order k , and satisfies $0 < \lambda_k \leq |f^{(k)}(x)| \leq h\lambda_k$. Then, there exists an absolute positive constant A such that*

$$\left| \sum_{a < n \leq b} e(f(n)) \right| \leq A \left(h^{\frac{2}{K}} (b-a) \lambda_k^{\frac{1}{2(K-1)}} + (b-a)^{1-\frac{2}{K}} \lambda_k^{\frac{1}{2(K-1)}} \right). \quad (23)$$

Proof. Without loss of generality, assume $A \geq 2$. When $\lambda_k \leq (b-a)^{\frac{4}{K-1}}$ or $\lambda_k \geq 1$, the conclusion holds trivially. Thus we may assume $(b-a)^{\frac{4}{K-1}} < \lambda_k < 1$. We prove by induction. When $k = 2$, the conclusion follows from Theorems 2.3. Assume the conclusion holds for $k-1 \geq 2$. We now prove that the conclusion also holds for k . Let l be a positive integer to be determined, $l \leq b-a$. For integers r , $1 \leq r \leq l-1$, let $J(x) = f(x+r) - f(x)$, $a < x \leq b-r$. From condition and the mean value theorem, we know

$$0 < r\lambda_k \leq |J^{(k-1)}(x)| \leq h(r\lambda_k), \quad a < x \leq b-r. \quad (24)$$

Thus, for the function $J(x)$, $a < x \leq b-r$, by the induction hypothesis, we know

$$\left| \sum_{a < n \leq b-r} e(J(n)) \right| \leq A \left(h^{\frac{4}{K}} (b-a) (r\lambda_k)^{\frac{1}{K-2}} + (b-a)^{1-\frac{4}{K}} (r\lambda_k)^{-\frac{1}{K-2}} \right). \quad (25)$$

From (25) and (19), we get:

$$\left| \sum_{a < n \leq b} e(f(n)) \right| \leq 2 \cdot \frac{b-a}{\sqrt{l}} + 2\sqrt{\frac{b-a}{l}} \sqrt{A} \cdot \left\{ \sum_{r=1}^{l-1} \left(h^{\frac{4}{K}} (b-a) (r\lambda_k)^{\frac{1}{K-2}} + (b-a)^{1-\frac{4}{K}} (r\lambda_k)^{-\frac{1}{K-2}} \right) \right\}^{\frac{1}{2}}. \quad (26)$$

Since $k \geq 3$, $K \geq 4$, so

$$\sum_{r=1}^{l-1} r^{\frac{1}{K-2}} < \int_0^l u^{\frac{1}{K-2}} du \leq 2l^{1-\frac{1}{K-2}}, \quad \sum_{r=1}^{l-1} r^{-\frac{1}{K-2}} < l^{1+\frac{1}{K-2}}. \quad (27)$$

From these three formulas, we get

$$\left| \sum_{a < n \leq b} e(f(n)) \right| \leq 2\frac{b-a}{\sqrt{l}} + 2\sqrt{A}(b-a)h^{\frac{2}{K}}(l\lambda_k)^{\frac{1}{2(K-2)}} + 2\sqrt{2A}(b-a)^{1-\frac{2}{K}}(l\lambda_k)^{-\frac{1}{2(K-2)}}. \quad (28)$$

Take $l = \lfloor \lambda_k^{-\frac{1}{K-1}} \rfloor$, From the assumption $k \geq 3$, $(b-a)^{\frac{4}{K-1}} < \lambda_k < 1$, we know

$$1 \leq l \leq (b-a)^{\frac{4}{K}} \leq (b-a), \quad (29)$$

satisfying the conditions of Lemma 3.1. Hence, from (27) and $\frac{1}{2}\lambda_k^{-\frac{1}{K-1}} < l \leq \lambda_k^{-\frac{1}{K-1}}$, it follows that

$$\left| \sum_{a < n \leq b} e(f(n)) \right| \leq 2\sqrt{2}(b-a)\lambda_k^{\frac{1}{2(K-1)}} + 2\sqrt{A}h^{\frac{2}{K}}(b-a)\lambda_k^{\frac{1}{2(K-1)}} + 4\sqrt{A}(b-a)^{1-\frac{2}{K}}\lambda_k^{-\frac{1}{2(K-1)}}. \quad (30)$$

Taking $A = 20$, it follows that (23) also holds for k . □

As a specific application, we use Theorem 4.1 to estimate $\sum_{a < n \leq b} n^{it}$.

Theorem 4.2. *Suppose $1 \leq a < b \leq 2a$, $|t| \geq 2$, integer $k \geq 2$, and $K = 2^{k-1}$. We have*

$$\sum_{a < n \leq b} n^{it} \ll a^{1-\frac{k}{2K-2}} |t|^{\frac{1}{2K-2}} + a^{1-\frac{2}{K}+\frac{k}{2K-2}} |t|^{-\frac{1}{2K-2}}. \quad (31)$$

Proof. In Theorem 4.1, take $f(x) = \frac{t}{2\pi} \log x$, then

$$\lambda_k = \frac{(k-1)!|t|}{2\pi(2a)^k} \leq |f^{(k)}(x)| \leq 2^k \lambda_k, \quad a < x \leq 2a. \quad (32)$$

From this and (23) the desired result follows. $\square$

Let us further discuss estimate (31). It is easy to see that in order for it be effective (ignoring the constant $\ll$) we need to satisfy the condition $|t|^{\frac{1}{k}} \leq a \leq |t|^{\frac{1}{k-4+4K-1}}$. (When $k=2$, condition becomes $|t|^{\frac{1}{2}} \leq a < \infty$.) In applications, $|t|$ and a are fixed. The problem is to determine k such that the estimate in (31) is optimal. To this end, let $l \geq 2$, $L = 2^{l-1}$, and $t_l = |t|^{\frac{1}{l-2+2L-1}}$. It can be verified directly: When $t_{l+1} \leq a \leq t_l$ in Theorem 4.2, taking $k=l$, the resulting estimate (31) is optimal.

Furthermore, it is easy to see that when $a \leq t_k$, the second term on the right-hand side of (31) is smaller than the first term and can be discarded; when $a \geq t_k$, the first term can be discarded.

The proof is completely identical to that of Theorem 4.2.

Theorem 4.3. Suppose $0 \leq \theta \leq 1$. Then, under the conditions of Theorem 2, we have

$$\sum_{a < n \leq b} (n + \theta)^{it} \ll a^{1 - \frac{k}{2K-2}} |t|^{\frac{1}{2K-2}} + a^{1 - \frac{2}{K} + \frac{k}{2K-2}} |t|^{-\frac{1}{2K-2}}. \quad (33)$$

5 Inversion Formulas

Under stronger conditions, instead of the upper bound estimate of Lemma 2.1, we can obtain an asymptotic formula for the exponential integral (Lemma 5.1). Therefore, replacing Theorems 2.3 we can also obtain an asymptotic formula for the Weyl sum (Theorem 5.2). The main term is also a Weyl sum, so it is usually called the inversion formula for Weyl sums.

Lemma 5.1. Suppose $F(x)$ is four times continuously differentiable on the interval $[a, b]$, and there exist positive constants λ_2, λ_3 such that when $a \leq x \leq b$, we have

$$0 < \lambda_2 \leq |F''(x)| \leq h\lambda_2, \quad F^{(3)}(x) \ll \lambda_3, \quad F^{(4)}(x) \ll \lambda_2^{-1} \lambda_3^2 = \lambda_4. \quad (34)$$

Furthermore, let there exist $a \leq c \leq b$ such that $F'(c) = 0$. Then we have

$$\begin{aligned} \int_a^b e^{iF(x)} dx &= \sqrt{2\pi} e^{i(\frac{\pi}{4} + F(c))} |F''(c)|^{-\frac{1}{2}} + O(\lambda_2^{-1} \lambda_3^{\frac{1}{3}}) \\ &\quad + O(\min(\lambda_3^{-\frac{1}{3}}, \lambda_2^{-\frac{1}{2}}, h|F'(a)|^{-1})) + O(\min(\lambda_3^{-\frac{1}{3}}, \lambda_2^{-\frac{1}{2}}, h|F'(b)|^{-1})), \end{aligned} \quad (35)$$

where the sign is “+” when $F''(x) > 0$ and “−” when $F''(x) < 0$.

Proof. Without loss of generality, assume $F''(x) < 0$. Then $F'(x)$ is decreasing, and c is the unique zero point of $F'(x)$. Let δ be a positive constant to be determined satisfying $0 < \delta \leq \min(c-a, b-c)$. We have

$$\int_a^b e^{iF(x)} dx = \int_a^{c-\delta} e^{iF(x)} dx + \int_{c-\delta}^{c+\delta} e^{iF(x)} dx + \int_{c+\delta}^b e^{iF(x)} dx. \quad (36)$$

From Lemma 1.3, we know

$$\int_{c+\delta}^b e^{iF(x)} dx \ll |F'(c+\delta)|^{-1} = \left| \int_c^{c+\delta} F''(x) dx \right|^{-1} \leq (\delta \lambda_2)^{-1}. \quad (37)$$

Similarly, we obtain $\int_a^{c-\delta} e^{iF(x)} dx \ll (\delta \lambda_2)^{-1}$. Now let us discuss the second integral on the right-hand side of (36). Since $F'(c) = 0$ and from Taylor's formula there exists θ , $|\theta| \leq 1$, such that when $|y| \leq \delta$, we have

$$F(y+c) = F(c) + \frac{1}{2} y^2 F''(c) + \frac{1}{6} y^3 F^{(3)}(c) + \frac{1}{24} y^4 F^{(4)}(c + \theta y). \quad (38)$$

From this and we get

$$\begin{aligned} \int_{c-\delta}^{c+\delta} e^{iF(x)} dx &= \int_{-\delta}^{\delta} \exp \left\{ i \left(F(c) + \frac{y^2}{2} F''(c) + \frac{y^3}{6} F'''(c) + O(\lambda_4 y^4) \right) \right\} dy \\ &= e^{iF(c)} \int_{-\delta}^{\delta} \exp \left\{ i \left(\frac{y^2}{2} F''(c) + \frac{y^3}{6} F'''(c) \right) \right\} dy + O(\delta^5 \lambda_4). \end{aligned} \quad (39)$$

Now we calculate the integral on the right-hand side of the above formula. Expanding e^x into a power series we get that this integral equals

$$\int_{-\delta}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{iy^3}{6}F'''(c)\right)^n dy = 2 \int_0^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) \left\{1 + \sum_{m=1}^{\infty} \frac{1}{(2m)!} \left(\frac{iy^3}{6}F'''(c)\right)^{2m}\right\} dy. \quad (40)$$

From Lemma 2.1, for any positive constant A , we have

$$\int_0^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) dy = \int_0^A \exp\left(\frac{iy^2}{2}F''(c)\right) dy + O((\delta\lambda_2)^{-1}), \quad (41)$$

where the constant in O is independent of A . From this and the well-known integral

$$\int_0^{\infty} \exp\left(\frac{iF''(c)}{2}y^2\right) dy = \frac{\sqrt{2}}{\sqrt{|F''(c)|}} \int_0^{\infty} \exp(-iu^2) du = \frac{\sqrt{2\pi}}{2\sqrt{|F''(c)|}} e^{-i\frac{\pi}{4}}, \quad (42)$$

we get

$$\int_0^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) dy = \frac{\sqrt{2\pi}}{2\sqrt{|F''(c)|}} e^{-i\frac{\pi}{4}} + O((\delta\lambda_2)^{-1}). \quad (43)$$

To estimate the other integrals on the right-hand side of (40), setting $y^2 = v$, and applying Lemma 1.4, we get

$$\begin{aligned} & \int_0^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) \sum_{m=1}^{\infty} \frac{1}{(2m)!} \left(\frac{iy^3}{6}F'''(c)\right)^{2m} dy \\ &= \frac{1}{2} \sum_{m=1}^{\infty} \frac{1}{(2m)!} \left(\frac{iF'''(c)}{6}\right)^{2m} \int_0^{\delta^2} \exp\left(\frac{iF''(c)}{2}v\right) v^{3m-\frac{1}{2}} dv \\ &\ll \sum_{m=1}^{\infty} \frac{1}{(2m)!} \left|\frac{F'''(c)}{6}\right|^{2m} \frac{\delta^{6m-1}}{|F''(c)|} \\ &\ll (\delta\lambda_2)^{-1} \sum_{m=1}^{\infty} \frac{1}{(2m)!} \left|\frac{F'''(c)}{6}\delta^3\right|^{2m} \\ &\ll (\delta\lambda_2)^{-1} \exp(c\delta^3\lambda_3), \end{aligned} \quad (44)$$

where c is a positive constant. Combining the above results, we obtain

$$\int_a^b e^{iF(x)} dx = \sqrt{\frac{2\pi}{|F''(c)|}} e^{-i\frac{\pi}{4}+iF(c)} + O((\delta\lambda_2)^{-1}) + O(\delta^5\lambda_4) + O((\delta\lambda_2)^{-1} \exp(c\delta^3\lambda_3)). \quad (45)$$

Now take $\delta = \lambda_3^{-\frac{1}{3}}$. If $\lambda_3^{-\frac{1}{3}} \leq \min(c-a, b-c)$ holds, then (35) follows from (45) (without the last two large O terms). If not hold, then either of the following inequalities holds: $a \leq c < a + \lambda_3^{-\frac{1}{3}}$ and $b - \lambda_3^{-\frac{1}{3}} < c \leq b$. At least one of them holds (This includes the cases $c = a$ or $c = b$). The discussions for the various possible situations are the same. First, assume the latter holds while the former does not. Then $a \leq c - \delta$, $b < c + \delta$, where δ still takes the value $\lambda_3^{-\frac{1}{3}}$. Thus, we have

$$\int_a^b e^{iF(x)} dx = \int_a^{c-\delta} e^{iF(x)} dx + \int_{c-\delta}^b e^{iF(x)} dx. \quad (46)$$

Similarly to the derivation of (39), we can obtain

$$\begin{aligned} \int_{c-\delta}^b e^{iF(x)} dx &= e^{iF(c)} \int_{-\delta}^{b-c} \exp\left\{i\left(\frac{y^2}{2}F''(c) + \frac{y^3}{6}F'''(c)\right)\right\} dy + O(\delta^5\lambda_4) \\ &= e^{iF(c)} \int_{-\delta}^{\delta} \exp\left\{i\left(\frac{y^2}{2}F''(c) + \frac{y^3}{6}F'''(c)\right)\right\} dy + I + O(\delta^5\lambda_4), \end{aligned} \quad (47)$$

where

$$\begin{aligned} I &= -e^{iF(c)} \int_{b-c}^{\delta} \exp\left\{i\left(\frac{y^2}{2}F''(c) + \frac{y^3}{6}F'''(c)\right)\right\} dy \\ &= -e^{iF(c)} \int_{b-c}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) dy \\ &\quad - e^{iF(c)} \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{iF'''(c)}{6}\right)^n \int_{b-c}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) y^{3n} dy. \end{aligned} \quad (48)$$

From Lemma 1.3, we can get

$$\begin{aligned} \int_{b-c}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) dy &\ll |F''(c)|^{-1}(b-c)^{-1} \\ &\ll \lambda_2^{-1}(b-c)^{-1} = \lambda_2^{-1} \left| \frac{F'(b) - F'(c)}{b-c} \right| |F'(b)|^{-1} \\ &\ll h|F'(b)|^{-1}. \end{aligned}$$

On the other hand, from Lemma 2.1, we get

$$\int_{b-c}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) dy \ll \lambda_2^{-\frac{1}{2}}.$$

Hence, we have

$$\int_{b-c}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) dy \ll \min\left(\delta, \lambda_2^{-\frac{1}{2}}, h|F'(b)|^{-1}\right). \quad (49)$$

Furthermore, from Lemma 1.3, we have uniformly for $n \geq 1$:

$$\int_{b-c}^{\delta} \exp\left(\frac{iy^2}{2}F''(c)\right) y^{3n} dy \ll \delta^{3n-1} \lambda_2^{-1}. \quad (50)$$

From (48), (49), and (50) we get

$$\begin{aligned} I &\ll \min\left(\delta, \lambda_2^{-\frac{1}{2}}, h|F'(b)|^{-1}\right) + \sum_{n=1}^{\infty} (\delta \lambda_2)^{-1} \frac{1}{n!} \left| \frac{\delta^3 F'''(c)}{6} \right|^n \\ &\ll \min\left(\delta, \lambda_2^{-\frac{1}{2}}, h|F'(b)|^{-1}\right) + (\delta \lambda_2)^{-1} \exp(c\delta^3 \lambda_3). \end{aligned} \quad (51)$$

Combining (46), (47), and (51), along with the discussion of the first integral on the right-hand side of (47), we deduce that (35) also holds (and the second large O does not appear).

Similarly, it can be proved that when the former holds while the latter does not, (35) also holds (and the third large O does not appear). $\square$

With Lemma 5.1, we can now prove the following inversion formula.

Theorem 5.2. *Assume on the interval $[a, b]$, $F(x) = 2\pi f(x)$ satisfies the conditions of Lemma 5.1, and $f'(x)$ is decreasing. Let $f'(a) = \beta$ and $f'(b) = \alpha$. Furthermore, for the integer v satisfying $\alpha < v \leq \beta$, define $f'(x_v) = v$. Then, we have*

$$\begin{aligned} \sum_{a < n \leq b} e(f(n)) &= e\left(-\frac{1}{8}\right) \sum_{a < v \leq b} |f''(x_v)|^{-\frac{1}{2}} e(f(x_v) - vx_v) \\ &\quad + O(\lambda_2^{-\frac{1}{2}}) + O(h(b-a)\lambda_3^{\frac{1}{3}}) + O(h \log(h(b-a)\lambda_2 + 2)). \end{aligned} \quad (52)$$

Proof. From (4) of Theorem 1.5 (taking $\eta = \frac{1}{2}$) we get

$$\begin{aligned} \sum_{a < n \leq b} e(f(n)) &= \sum_{\alpha - \frac{1}{2} < v < \beta + \frac{1}{2}} \int_a^b e(f(x) - vx) dx + O(\log(\beta - \alpha + 2)) \\ &= \sum_{a+1 < v < \beta-1} \int_a^b e(f(x) - vx) dx + O(\lambda_2^{-\frac{1}{2}}) + O(\log(\beta - \alpha + 2)), \end{aligned} \quad (53)$$

where the last step used Lemma 2.1. From Lemma 5.1 we know

$$\begin{aligned} \sum_{a+1 < v < \beta-1} \int_a^b e(f(x) - vx) dx &= e\left(-\frac{1}{8}\right) \sum_{a+1 < v < \beta-1} |f''(x_v)|^{-\frac{1}{2}} e(f(x_v) - vx_v) \\ &\quad + O\left(\sum_{a+1 < v < \beta-1} \lambda_2^{-1} \lambda_3^{\frac{1}{3}}\right) + O\left(h \sum_{a+1 < v < \beta-1} ((v-\alpha)^{-1} + (\beta-v)^{-1})\right) \\ &= e\left(-\frac{1}{8}\right) \sum_{a < v < \beta} |f''(x_v)|^{-\frac{1}{2}} e(f(x_v) - vx_v) \\ &\quad + O(\lambda_2^{-\frac{1}{2}}) + O((\beta - \alpha)\lambda_2^{-1} \lambda_3^{\frac{1}{3}}) + O(h \log(\beta - \alpha + 2)). \end{aligned} \quad (54)$$

Noting that $\beta - \alpha = f'(a) - f'(b) \leq h\lambda_2(b-a)$, (52) follows from (53) and (54). $\square$

For the case where $f'(x)$ is increasing, it is not difficult to obtain the corresponding result:

$$\begin{aligned} \sum_{a < n \leq b} e(f(n)) = & e\left(\frac{1}{8}\right) \sum_{\beta < v < \alpha} |f''(x_v)|^{-\frac{1}{2}} e(f(x_v) - vx_v) \\ & + O(\lambda_2^{-\frac{1}{2}}) + O(h(b-a)\lambda_3^{\frac{1}{3}}) + O(h \log((b-a)\lambda_2 + 2)). \end{aligned} \quad (55)$$

6 Theory of Exponent Pairs

Using Theorem 3.1 or Theorem 5.2, an upper bound estimate of a given exponential sum can be transformed into an upper bound estimate of another exponential sum. Theorem 3.1 does not impose any conditions on $f(x)$ and is broadly applicable. Although Theorem 5.2 imposes strong conditions on $f(x)$, these conditions are satisfied in many important applications. The current problem is how to combine these two to obtain the best possible upper bound estimate. This is precisely the topic of this section.

In this section we consider exponential sums

$$S = \sum_{a < n \leq a+d} e(f(n)), \quad 1 < d \leq a, \quad (56)$$

where $f(x)$ has at least 5 continuous derivatives on the interval $[a, 2a]$ and satisfies

$$\lambda_1 a^{1-j} \ll |f^{(j)}(x)| \ll \lambda_1 a^{1-j}, \quad j = 1, 2, 3, 4, \quad \lambda_1 > \frac{1}{2}. \quad (57)$$

For $j = 4$, only the upper bound estimate is required to hold.

If there exists a pair of non-negative real numbers (κ, λ) : $0 \leq \kappa \leq \frac{1}{2} \leq \lambda \leq 1$ such that the estimate $S \ll \lambda_1^\kappa a^\lambda$ holds, then (κ, λ) is called an exponent pair. The constant implied by $\ll$ here is related only to the constants in condition (57).

Lemma 6.1. (a) $(0, 1)$ is an exponent pair; (b) the set of exponent pairs forms a convex set; (c) $(\frac{1}{2}, \frac{1}{2})$ is an exponent pair.

Proof. (a) is obvious.

Suppose (κ, λ) and (κ', λ') are two exponent pairs. Then for any $0 \leq t \leq 1$, we have

$$S = S^t S^{1-t} \ll (\lambda_1^\kappa a^\lambda)^t (\lambda_1^{\kappa'} a^{\lambda'})^{1-t} = \lambda_1^{\kappa t + \kappa'(1-t)} a^{\lambda t + \lambda'(1-t)}. \quad (58)$$

Thus, for any $0 \leq t \leq 1$,

$$(\kappa t + \kappa'(1-t), \lambda t + \lambda'(1-t)) \quad (59)$$

is also an exponent pair, which proves (b).

When (57) (for $j \leq 2$) is satisfied, we deduce from Theorem 2.3:

$$S \ll a(\lambda_1 a^{-1})^{\frac{1}{2}} + (\lambda_1 a^{-1})^{-\frac{1}{2}} \ll \lambda_1^{\frac{1}{2}} a^{\frac{1}{2}}, \quad (60)$$

which proves (c). □

From Lemma 6.1, it is immediately clear that starting from the exponent pairs $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$, repeatedly applying (7) yields infinitely many exponent pairs. We denote the set of all such exponent pairs by $\mathcal{C}_1$.

The following lemma allows us to obtain new exponent pairs from a given exponent pair using Theorem 3.1.

Lemma 6.2. If (κ, λ) is an exponent pair, then

$$(\kappa', \lambda') = (\kappa(2\kappa + 2)^{-1}, \frac{1}{2} + \lambda(2\kappa + 2)^{-1}) \quad (61)$$

is also an exponent pair. Here (57) only needs to hold for $1 \leq j \leq 3$

Proof. First, from assumption it is clear that $0 \leq \kappa' \leq \frac{1}{2} \leq \lambda' \leq 1$. Let l be a positive integer to be determined. From Theorem 3.1, we can obtain

$$S^2 \ll a^2 l^{-1} + a l^{-1} \sum_{r=1}^{l-1} \left| \sum_{a < n \leq a+d-r} e(f(n+r) - f(n)) \right| + l^2. \quad (62)$$

In fact when $l \leq d$, (62) follows from Theorem 3.1; when $l > d$, (62) is trivially true due to the added l^2 term. For a fixed r , let $g_r(n) = g(n) = f(n+r) - f(n)$. We have

$$g^{(j)}(x) = rf^{(j+1)}(x + \theta r), \quad j \geq 0, |\theta| \leq 1. \quad (63)$$

Thus, from (57), when $a < x \leq a + d - r$, $1 < d \leq a$, we have

$$r\lambda_1 a^{-j} \ll g^{(j)}(x) \ll r\lambda_1 a^{-j}, \quad j \geq 0. \quad (64)$$

If $\lambda_1 \leq a^{\frac{1}{2}}$, from $(\frac{1}{2}, \frac{1}{2})$ being an exponent pair and $\lambda \geq \frac{1}{2}$, we get

$$S \ll \lambda_1^{\frac{1}{2}} a^{\frac{1}{2}} = \lambda_1^{\frac{1}{2}} a^{\frac{1}{2}} + \lambda_1^{\frac{\lambda}{2\kappa+2}} a^{-\frac{\lambda}{2\kappa+2}} \ll \lambda_1^{\frac{1}{2}} \lambda_1^{-\frac{\kappa}{\kappa+1}} a^{\lambda'} \ll \lambda_1^{\kappa'} a^{\lambda'}, \quad (65)$$

so Lemma holds. $\square$

Therefore, we may assume $\lambda_1 > a^{\frac{1}{2}}$. Let c be a certain positive constant. When $r < ca\lambda_1^{-1}$, from (64) ($j = 2$) and Theorem 2.3 (since $\lambda_1 > a^{1/2}$, the first term on the right-hand side of (18) can be ignored), we know

$$al^{-1} \sum_{r < ca\lambda_1^{-1}} \left| \sum_{a < n \leq a+d-r} g_r(n) \right| \ll al^{-1} \sum_{r < ca\lambda_1^{-1}} (r\lambda_1 a^{-2})^{\frac{1}{2}} \ll a^2 l^{-1} \lambda_1^{-\frac{1}{2}} \sum_{r < ca\lambda_1^{-1}} r^{-\frac{1}{2}} \ll a^2 l^{-1}, \quad (66)$$

where the constant implied by $\ll$ depends on c , and we used $\lambda_1 > a^{\frac{1}{2}}$. For the remaining r , using (κ, λ) as an exponent pair and (64) (In (57), λ_1, a should be replaced here by $c^{-1}r\lambda_1 a^{-1}$ and a . From the restriction on r here $c^{-1}r\lambda_1 a^{-1} \geq 1$, so $\lambda_1 > \frac{1}{2}$ is satisfied), we obtain

$$al^{-1} \sum_{ca\lambda_1^{-1} \leq r \leq l-1} \left| \sum_{a < n \leq a+d-r} g_r(n) \right| \ll al^{-1} \sum_{r \leq l-1} (r\lambda_1 a^{-1})^{\kappa} a^{\lambda} \ll \lambda_1^{\kappa} a^{1-\kappa+\lambda} l^{\kappa}. \quad (67)$$

Substituting the above two formulas into (62) yields $S^2 \ll a^2 l^{-1} + \lambda_1^{\kappa} a^{1-\kappa+\lambda} l^{\kappa} + l^2$. Now take $l = \left[a^{1-\frac{\lambda}{\kappa+1}}, \lambda_1^{-\frac{\kappa}{\kappa+1}} \right]$. If such a choice of l satisfies $l \geq 1$, then we obtain

$$S^2 \ll \lambda_1^{\frac{\kappa}{\kappa+1}} a^{1+\frac{\lambda}{\kappa+1}} + \lambda_1^{-\frac{2\kappa}{\kappa+1}} a^{2-\frac{2\lambda}{\kappa+1}}. \quad (68)$$

Furthermore, $1 + \frac{\lambda}{\kappa+1} \geq 2 - \frac{2\lambda}{\kappa+1}$. From this and $\lambda_1 > \frac{1}{2}$, the second term on the right-hand side of (68) can be ignored, i.e., $S^2 \ll \lambda_1^{\frac{\kappa}{\kappa+1}} a^{1+\frac{\lambda}{\kappa+1}}$. This proves the desired result. If such a choice gives $l = 0$, then there must be $0 < a^{1-\frac{\lambda}{\kappa+1}} \lambda_1^{-\frac{\kappa}{\kappa+1}} < 1$, i.e., $\lambda_1^{\frac{\kappa}{\kappa+1}} a^{1+\frac{\lambda}{\kappa+1}} > a^2$. Thus, it clearly holds. Lemma 6.2 is proved. $\square$

We denote the set of all exponent pairs obtained from $\mathcal{C}_1$ by repeatedly applying convexity and Lemma 6.2 as $\mathcal{C}_2$. We now prove that when $(\kappa, \lambda) \in \mathcal{C}_2$, we must have $\kappa + 2\lambda \geq \frac{3}{2}$. Because the new exponent pairs formed by convexity of two exponent pairs with this property clearly still satisfy it, those belonging to $\mathcal{C}_1$ and the exponent pairs obtained from $\mathcal{C}_1$ by convexity all satisfy it. On the other hand, for the exponent pair (κ', λ') obtained from Lemma 6.2, we have

$$\kappa' + 2\lambda' = \frac{\kappa}{2\kappa+2} + 2 \left(\frac{1}{2} + \frac{\lambda}{2\kappa+2} \right) = 1 + \frac{\kappa+2\lambda}{2\kappa+2} \geq 1 + \frac{\kappa+1}{2\kappa+2} = \frac{3}{2}, \quad (69)$$

Next, we use Theorem 5.2 to construct new exponent pairs.

Lemma 6.3. *Let (κ, λ) be an exponent pair satisfying condition (18). Then $(\kappa', \lambda') = \left(\lambda - \frac{1}{2}, \kappa + \frac{1}{2} \right)$ is also an exponent pair.*

Proof. It is clear that the conditions $0 \leq \kappa' \leq \frac{1}{2} \leq \lambda' \leq 1$ are satisfied. Without loss of generality, assume $f'(x) < 0$. In Theorem 5.2, take $b = a + d$, $\lambda_2 = \lambda_1 a^{-1}$, $\lambda_3 = \lambda_1 a^{-2}$. From (57) (for $j \leq 3$, and the upper bound estimate for $j = 4$), we know that the conditions (34) of Lemma 5.1 are satisfied. Therefore, from (52), we obtain

$$S = e \left(-\frac{1}{8} \right) \sum_{\alpha < v \leq \beta} |f''(x_v)|^{-\frac{1}{2}} e(f(x_v) - vx_v) + O(\lambda_1^{-\frac{1}{2}} a^{\frac{1}{2}}) + O(\log(\lambda_1 + 2)) + O(\lambda_1^{\frac{1}{3}} a^{\frac{1}{3}}). \quad (70)$$

Next we estimate the sum on the right-hand side, denoting it by S_1 . Let $g(v) = f(x_v) - vx_v$, then

$$S_1 = \sum_{\alpha < v \leq \beta} |f''(x_v)|^{-\frac{1}{2}} e(g(v)). \quad (71)$$

Let $x = h(y)$ be the inverse function of $y = f'(x)$. Thus, we have $g(y) = f(h(y)) - yh(y)$, $\alpha \leq y \leq \beta$. Let us compute the derivatives of $g(y)$.

$$\begin{aligned} g'(y) &= f'(h(y))h'(y) - h(y) - yh'(y) = -h(y), \\ g''(y) &= -h'(y) = -\frac{dx}{dy} = -\left[\left(\frac{dy}{dx}\right)^{-1}\right]_{x=h(y)} = -(f''(h(y)))^{-1}. \end{aligned}$$

From this, we know

$$g'(y)f'(h(y)) = g'(y)f'(-g'(y)) = -1.$$

Furthermore,

$$g''(y)f'(h(y)) + g'(y)f''(h(y))(f'(h(y)))^{-1} = 0, \quad g''(y) = f''(h(y))(f'(h(y)))^{-3}. \quad (72)$$

From $a \leq h(y) \leq 2a$ and (57) (for $1 \leq j \leq 3$) we know

$$a\lambda_1^{1-j} \ll |g^{(j)}(y)| \ll a\lambda_1^{1-j}, \quad j = 1, 2, 3. \quad (73)$$

Using Leibniz's rule, we can further compute the derivatives of $g(y)$. Using the upper bound estimate of (57) for $j = 4$, we obtain $g^{(4)}(y) \ll a\lambda_1^{-3}$. Using summation by parts, since (κ, λ) is an exponent pair, along with (57), (73), we obtain

$$S_1 \ll \lambda_1^{-\frac{1}{2}} a^{\frac{1}{2}} a^\kappa \lambda_1^{-\kappa} = \lambda_1^{-\kappa - \frac{1}{2}} a^{\kappa + \frac{1}{2}}. \quad (74)$$

From this and (70), we obtain $S \ll \lambda_1^{\kappa'} a^{\lambda'} + \lambda_1^{\frac{1}{3}} a^{\frac{1}{3}}$. Noting that $\lambda' \geq \frac{1}{2}$, if $\kappa' \geq \frac{1}{3}$, then the conclusion follows from the above formula. If $\kappa' < \frac{1}{3}$, then

(i) When $a \geq \lambda_1^{(1-3\kappa')(3\lambda'-1)^{-1}}$, we have $a^{\frac{1}{3}} \lambda_1^{\frac{1}{2}} = \lambda_1^{\kappa'} a^{\frac{1}{3}} \lambda_1^{\frac{1}{3}-\kappa'} \ll \lambda_1^{\kappa'} a^{\lambda'}$.

(ii) When $a < \lambda_1^{(1-3\kappa')(3\lambda'-1)^{-1}}$, using $2\kappa' + \lambda' \geq 1$ we obtain

$$S \ll a = a^{\kappa'} a^{1-\lambda'} < a^{\kappa'} \lambda_1^{(1-3\kappa')(1-\lambda')(3\lambda'-1)^{-1}} \leq \lambda_1^{\kappa'} a^{\lambda'}.$$

Thus, the conclusion holds in all cases. $\square$

Theorem 6.4. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}_t$ ($0 \leq t \leq 1$) be the transformations on real pairs defined as follows:

$$\mathcal{A}((\kappa, \lambda)) = \left(\kappa(2\kappa + 2)^{-1}, \frac{1}{2} + \lambda(2\kappa + 2)^{-1} \right), \quad (75)$$

$$\mathcal{B}((\kappa, \lambda)) = \left(\lambda - \frac{1}{2}, \kappa + \frac{1}{2} \right), \quad (76)$$

$$\mathcal{C}_t((\kappa, \lambda), (\kappa', \lambda')) = (\kappa t + \kappa'(1-t), \lambda t + \lambda'(1-t)). \quad (77)$$

Furthermore, let $\mathcal{E}$ be the set of real pairs (κ, λ) formed as follows: $(0, 1) \in \mathcal{E}$, and any real pair obtained from $(0, 1)$ through a finite number of transformations (75), (76), and (77) belongs to $\mathcal{E}$. Thus, every real pair (κ, λ) in $\mathcal{E}$ is an exponential pair.

The theory of exponent pairs began with the research of van der Corput and E. Phillips. Although more than 60 years have passed since then, no other exponential pairs besides the set $\mathcal{E}$ have been found.

In practical applications, the complexity of the situation often obscures the best method for selecting an exponential pair — even when minimizing some function $F(k, \lambda)$. For a general F , this problem remains unsolved to this day. However, for $F(x, y) = x + y$, R. A. Rankin solved this problem. He proved that if we let $\alpha = 0.3290213568 \dots$, then for any arbitrarily small positive number ϵ , $(\kappa, \lambda) = \left(\frac{1}{2}\alpha + \epsilon, \frac{1}{2} + \frac{1}{2}\alpha + \epsilon \right)$ is the exponential pair in $\mathcal{E}$ that minimizes $\kappa + \lambda$ (in the sense of accuracy ϵ). The ten pairs of exponents listed above perfectly approximate this exponential pair sequentially.

II. Vinogradov Method

In 1934, I. M. Vinogradov proposed a new method for estimating Weyl sums. Loo-keng Hua pointed out that the key to his method lies in estimating the mean value of exponential sums of the form (78). Estimates for individual exponential sums are then obtained from the mean value estimates of these exponential sums. This mean value estimate (Theorem 7.4) is commonly known as the Vinogradov Mean Value Theorem. This method has been successfully applied to the Waring problem and the non-zero region of the ζ function, yielding the best results to date in these areas.

7 Mean Value Estimates of Exponential Sums

Let k, n, P be positive integers. The mean value of the exponential sum is

$$J_k^{(n)}(P) = \int_0^1 \cdots \int_0^1 \left| \sum_{1 \leq x \leq P} e(\alpha_n x^n + \cdots + \alpha_1 x) \right|^{2k} d\alpha_1 \cdots d\alpha_n. \quad (78)$$

The mean value $J_k^{(n)}(P)$ has a clear arithmetic interpretation. Generally speaking, let $\lambda_1, \dots, \lambda_n$ be integers. It is easy to see that the integral

$$J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) = \int_0^1 \cdots \int_0^1 \left| \sum_{1 \leq x \leq P} e(\alpha_n x^n + \cdots + \alpha_1 x) \right|^{2k} \cdot e(-(\alpha_n \lambda_n + \cdots + \alpha_1 \lambda_1)) d\alpha_1 \cdots d\alpha_n \quad (79)$$

is equal to the number of solutions to the following indeterminate system of equations:

$$\begin{cases} \sum_{r=1}^k (x_r^l - y_r^l) = \lambda_l, & l = 1, \dots, n, \\ 1 \leq x_r \leq P, 1 \leq y_r \leq P, & r = 1, \dots, k. \end{cases} \quad (80)$$

In particular we have $J_k^{(n)}(P) = J_k^{(n)}(P; 0, \dots, 0)$.

Using the elementary properties of integrals and indeterminate equations, we can immediately infer that the number of solutions to the integral (79), i.e., the indeterminate system (80), has the following properties:

- (a) $J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) \geq 0$, $J_k^{(n)}(P) \geq P^k$. The former follows from the fact that the number of solutions to the indeterminate system must be non-negative; the latter holds because the solutions $1 \leq x_r = y_r \leq P$ for $r = 1, \dots, k$ necessarily satisfy the system (80) (with $\lambda_1 = \dots = \lambda_n = 0$).
- (b) $J_k^{(n)}(P; \lambda_1, \dots, \lambda_n)$ is an increasing function of P . This is because as the range of the variables expands, the number of solutions to the indeterminate system does not decrease.
- (c) $J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) \leq J_k^{(n)}(P)$. This follows by taking absolute values in the integrand of (79).
- (d) $J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) > 0$ implies the necessary condition for the system (80) to be solvable: $|\lambda_l| \leq k(P^l - 1)$, $l = 1, \dots, n$. This follows from the range of the variables in the indeterminate system (80).
- (e) $\sum_{\lambda_1, \dots, \lambda_n} J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) = P^{2k}$. Here the summation is over all possible values of $\lambda_1, \dots, \lambda_n$ (which, by (d), is a finite set). This is because for two different sets of values $(\lambda_1, \dots, \lambda_n)$ and $(\lambda'_1, \dots, \lambda'_n)$, the corresponding arbitrary solution sets $(x_1, \dots, x_k; y_1, \dots, y_k)$ and $(x'_1, \dots, x'_k; y'_1, \dots, y'_k)$ must be different. Therefore, the sum is equal to the number of arrays $(x_1, \dots, x_k; y_1, \dots, y_k)$, which is P^{2k} .
- (f) $J_k^{(n)}(P) > (2k)^{-n} P^{2k - \frac{1}{2}(n^2 + n)}$. From (e), (d), and (c), we have

$$P^{2k} = \sum_{\substack{|\lambda_l| \leq k(P^l - 1) \\ l=1, \dots, n}} J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) < J_k^{(n)}(P) \prod_{l=1}^n (2kP^l) = (2k)^n P^{\frac{1}{2}n(n+1)} J_k^{(n)}(P),$$

which yields the desired conclusion.

- (g) Let A be an integer. Then the number of solutions to the indeterminate system (80) (with $\lambda_1 = \dots = \lambda_n = 0$) is equal to the number of solutions to the indeterminate system

$$\begin{cases} \sum_{r=1}^k (\tilde{x}_r^l - \tilde{y}_r^l) = 0, & l = 1, \dots, n, \\ 1 + A \leq \tilde{x}_r \leq P + A, 1 + A \leq \tilde{y}_r \leq P + A, & r = 1, \dots, k. \end{cases} \quad (81)$$

This can be seen by applying the variable substitution $\tilde{x}_r = x_r + A$, $\tilde{y}_r = y_r + A$, $r = 1, \dots, k$. From this, it immediately follows: for any integer A ,

$$J_k^{(n)}(P) = \int_0^1 \cdots \int_0^1 \left| \sum_{1+A \leq x \leq P+A} e(\alpha_n x^n + \cdots + \alpha_1 x) \right|^{2k} d\alpha_1 \cdots d\alpha_n. \quad (82)$$

Let $p > n \geq 2$ be a prime, $k \geq n$. Let $J_{k,1}^{(n)}(P, p)$ denote the number of solutions to the indeterminate system (80) (with $\lambda_1 = \cdots = \lambda_n = 0$) such that among the arrays $x_1, \dots, x_k$ and $y_1, \dots, y_k$, there are at least n numbers in x that are pairwise incongruent modulo p , and at least n numbers in y that are also pairwise incongruent modulo p . Further, let

$$J_k^{(n)}(P) = J_{k,1}^{(n)}(P, p) + J_{k,2}^{(n)}(P, p). \quad (83)$$

Clearly, $J_{k,2}^{(n)}(P, p)$ is the number of solutions to the indeterminate system (80) (with $\lambda_1 = \cdots = \lambda_n = 0$) among the arrays $x_1, \dots, x_k$ and $y_1, \dots, y_k$, where at most $n-1$ pairs of x numbers are pairwise incongruent modulo p , and at most $n-1$ pairs of y numbers are pairwise incongruent modulo p . We estimate $J_{k,1}^{(n)}(P, p)$ and $J_{k,2}^{(n)}(P, p)$ separately. Both reduce to estimating $J_{k-n}^{(n)}(\lfloor P/p \rfloor + 1)$. Thus, from (83), we obtain a recursive estimate for the mean value $J_k^{(n)}(P)$, from which we obtain the upper bound estimate of $J_k^{(n)}(P)$.

Lemma 7.1. *Let n, m be positive integers, $p > n$ be a prime, and $h_1, \dots, h_n, A$ be integers. Let T be the number of solutions to the system of congruences*

$$\sum_{r=1}^n x_r^l \equiv h_l \pmod{p^l}, \quad l = 1, 2, \dots, n, \quad (84)$$

subject to the conditions

$$x_r \not\equiv x_{r'} \pmod{p}, \quad r \neq r', \quad \text{and } 1 + A \leq x_r \leq mp^n + A, \quad r = 1, \dots, n. \quad (85)$$

Then $T \leq n!m^n p^{\frac{1}{2}n(n-1)}$.

Proof. First, we can assume $A = 0$. It is easy to see that via the congruence relation $y_r \equiv x_r \pmod{p^n}$, $r = 1, \dots, n$, we can establish a one-to-one correspondence between the arrays $(x_1, \dots, x_n)$ satisfying (85) and the arrays $(y_1, \dots, y_n)$ satisfying the conditions

$$y_r \not\equiv y_{r'} \pmod{p}, \quad r \neq r', \quad 1 \leq y_r \leq mp^n, \quad r = 1, \dots, n. \quad (86)$$

Simultaneously, if $(x_1, \dots, x_n)$ satisfies the system, then the corresponding $(y_1, \dots, y_n)$ also satisfies, and vice versa. This proves that we may assume $A = 0$.

Secondly, every x_r satisfying (85) (with $A = 0$) can be uniquely expressed as

$$x_r = z_r + l_r p^n, \quad 1 \leq z_r \leq p^n, \quad 0 \leq l_r \leq m-1. \quad (87)$$

In this way, if $x_1, \dots, x_n$ satisfy (84) and (85), then their corresponding $z_1, \dots, z_n$ also satisfy. Moreover, if $z_1, \dots, z_n$ satisfy, then the array $x_1, \dots, x_n$ determined by (87) also satisfies (84) and (85) (with $A = 0$). Furthermore, there are m^n such arrays $x_1, \dots, x_n$. Therefore, if V denotes the number of solutions to (84) (with $A = 0$ and $m = 1$), then we have $T = m^n V$. Now let us estimate V . Let $\vec{g}$ denote the integer array $(g_1, \dots, g_n)$, and let $Q(\vec{g})$ denote the number of solutions to the following system of congruences:

$$\sum_{r=1}^n z_r^l \equiv g_l \pmod{p^n}, \quad l = 1, \dots, n, \quad 1 \leq z_r \leq p^n.$$

It is easy to see that $V = \sum_{\vec{g}} Q(\vec{g})$, where the summation is over all arrays $\vec{g}$ satisfying the following conditions:

$g_l \equiv h_l \pmod{p^l}$, $1 \leq g_l \leq p^n$, $l = 1, \dots, n$. For each fixed l , the number of g_l is obviously p^{n-1} . Therefore, the number of integer arrays $\vec{g}$ is $p^{n-1} \cdot p^{n-2} \cdots p^1 \cdot 1 = p^{\frac{1}{2}n(n-1)}$. Thus, $V \leq p^{\frac{1}{2}n(n-1)} \max_{\vec{g}} Q(\vec{g})$. Consequently, estimating V reduces to estimating $Q(\vec{g})$. We now prove that for any $\vec{g}$, we must have $Q(\vec{g}) \leq n!$. If the system of congruences has no solution, then it holds trivially. If it has a solution, let $z_1^0, \dots, z_n^0$ be one such solution, and set $F(u) = \prod_{r=1}^n (u - z_r^0)$.

Furthermore, let $z_1, \dots, z_n$ be any solution. Note that $p > n$. From the relationship between roots and coefficients of polynomials, and the properties of elementary symmetric polynomials, we can deduce

$$\prod_{r=1}^n (u - z_r) \equiv F(u) \pmod{p^n}. \quad (88)$$

Hence, for any z_s ($1 \leq s \leq n$), we must have

$$F(z_s) = \prod_{r=1}^n (z_s - z_r^0) \equiv 0 \pmod{p^n}.$$

There must exist one and only one $r = r(s)$ such that $z_r^0 \equiv z_s \pmod{p^n}$. From this and condition we have $z_r^0 = z_s$. This proves that any solution must be a rearrangement of $z_1^0, \dots, z_n^0$, and vice versa. The number of such arrangements is $n!$, i.e., $Q(\vec{g}) = n!$. Thus we obtain (11). $\square$

Now let us estimate $J_{k,1}^{(n)}(P, p)$.

Lemma 7.2. *Suppose $p > n \geq 2$ is a prime, and $k > n$. Let m be the smallest integer such that $mp^n \geq P$. Then we have*

$$J_{k,1}^{(n)}(P, p) \leq (C_n^k)^2 n! m^n p^{2k + \frac{1}{2}(n^2 - 5n)} P^n J_{k-n}^{(n)} \left(\left\lfloor \frac{P}{p} \right\rfloor + 1 \right). \quad (89)$$

Proof. For the indeterminate system (80), we perform the integer variable substitution

$$\begin{cases} x_r = u_r + pw_r, & 1 \leq u_r \leq p, \ 1 \leq u_r + pw_r \leq P, \ r = 1, \dots, k, \\ y_r = v_r + pz_r, & 1 \leq v_r \leq p, \ 1 \leq v_r + pz_r \leq P, \ r = 1, \dots, k. \end{cases} \quad (90)$$

Within the considered range of the variables, this substitution is bijective. Thus, the number of solutions to the indeterminate system (80) (with $\lambda_1 = \dots = \lambda_n = 0$), $J_k^{(n)}(P)$, is equal to the number of solutions to the following indeterminate system (with $4k$ variables):

$$\begin{cases} \sum_{r=1}^k ((u_r + pw_r)^l - (v_r + pz_r)^l) = 0, \ l = 1, \dots, n, \\ 1 \leq u_r \leq p, \ 1 \leq u_r + pw_r \leq P, \ r = 1, \dots, k, \\ 1 \leq v_r \leq p, \ 1 \leq v_r + pz_r \leq P, \ r = 1, \dots, k. \end{cases} \quad (91)$$

From the definition of $J_{k,1}^{(n)}(P, p)$, it is exactly the number of solutions to (91) such that among $u_1, \dots, u_k$ and $v_1, \dots, v_k$, there are at least n distinct numbers in each. Let T_1 denote the number of solutions to (91) such that $u_1, \dots, u_n$ are pairwise distinct, and $v_1, \dots, v_n$ are pairwise distinct. It is easy to see that $J_{k,1}^{(n)}(P, p) \leq (C_n^k)^2 T_1$. For a fixed u , $1 \leq u \leq p$, let

$$S(u) = \sum_{1 \leq u + pw \leq P} e(\alpha_n(u + pw)^n + \dots + \alpha_1(u + pw)). \quad (92)$$

Then we have

$$T_1 = \int_0^1 \dots \int_0^1 \left| \sum_{\substack{u_1=1 \\ u_r \neq u_s}}^p \dots \sum_{\substack{u_n=1 \\ r \neq s}}^p S(u_1) \dots S(u_n) \right|^2 \cdot \left| \sum_{u=1}^p S(u) \right|^{2k-2n} d\alpha_1 \dots d\alpha_n. \quad (1)$$

By Hölder's inequality we obtain

$$T_1 \leq p^{2k-2n-1} \sum_{u=1}^p \int_0^1 \dots \int_0^1 \left| \sum_{\substack{u_1=1 \\ u_r \neq u_s}}^p \dots \sum_{\substack{u_n=1 \\ r \neq s}}^p S(u_1) \dots S(u_n) \right|^2 |S(u)|^{2k-2n} d\alpha_1 \dots d\alpha_n. \quad (93)$$

The multiple integral on the right-hand side (with u as a parameter, $1 \leq u \leq p$) is the number of solutions to the following indeterminate system (with u as a parameter, $1 \leq u \leq p$):

$$\begin{cases} \sum_{r=1}^n ((u_r + pw_r)^l - (v_r + pz_r)^l) + \sum_{r=n+1}^k ((u + pw_r)^l - (u + pz_r)^l) = 0, \ l = 1, \dots, n, \\ 1 \leq u_r \leq p, \ 1 \leq u_r + pw_r \leq P, \ u_r \text{ pairwise distinct}, & r = 1, \dots, n, \\ 1 \leq v_r \leq p, \ 1 \leq v_r + pz_r \leq P, \ v_r \text{ pairwise distinct}, & r = 1, \dots, n, \\ 1 \leq u + pw_r \leq P, \ 1 \leq u + pz_r \leq P, & r = n+1, \dots, k. \end{cases} \quad (94)$$

Using the binomial expansion, it is easy to directly verify that (94) and the following system have the same number of solutions:

$$\begin{cases} \sum_{r=1}^n ((u_r - u + pw_r)^l - (v_r - u + pz_r)^l) + \sum_{r=n+1}^k p^l (w_r^l - z_r^l) = 0, \ l = 1, \dots, n, \\ u_r, w_r, v_r, z_r \text{ satisfy the same conditions as system (94)}. \end{cases} \quad (94')$$

It is easy to see that the $4n$ variables $u_r, w_r, v_r, z_r, r = 1, \dots, n$, of system (94) (i.e., (94')) must satisfy the system of congruences

$$\begin{cases} \sum_{r=1}^n ((u_r - u + pw_r)^l - (v_r - u + pz_r)^l) \equiv 0 \pmod{p^l}, & l = 1, \dots, n, \\ u_r, w_r, v_r, z_r, r = 1, \dots, n, \text{ satisfy the same conditions as system (94)}. \end{cases} \quad (95)$$

For any fixed $v_r, z_r, 1 \leq r \leq n$, let $h_l = \sum_{r=1}^n (v_r - u + pz_r)^l, l = 1, \dots, n$. It is easy to prove that the number of solutions to the system of congruences

$$\begin{cases} \sum_{r=1}^n (u_r - u + pw_r)^l \equiv h_l \pmod{p^l}, & l = 1, \dots, n, \\ 1 \leq u_r \leq p, \ u_r \text{ pairwise distinct}, \ 1 \leq u_r + pw_r \leq P, & r = 1, \dots, n \end{cases} \quad (96)$$

is equal to the number of solutions to

$$\begin{cases} \sum_{r=1}^n t_r^l \equiv h_l \pmod{p^l}, & l = 1, \dots, n, \\ -u + 1 \leq t_r \leq -u + P, \ t_r \text{ pairwise incongruent mod } p, & r = 1, \dots, n \end{cases} \quad (97)$$

this can be seen by establishing the correspondence $t_r = u_r - u + pw_r, r = 1, \dots, n$. By Lemma 7.1, the number of solutions to (97) is $\leq n!m^n p^{\frac{1}{2}n(n-1)}$. Since when v_r, z_r satisfy the conditions of (95), the number of arrays $h_1, \dots, h_n$ does not exceed P^n . Therefore, the number of solutions to the system of congruences (95) is $\leq n!m^n p^{\frac{1}{2}n(n-1)} P^n$.

Let $u_r, w_r, v_r, z_r, 1 \leq r \leq n$ be a solution to (95), and set $\lambda_l = -p^{-l} \sum_{r=1}^n ((u_r - u + pw_r)^l - (v_r - u + pz_r)^l)$. If these $4n$ numbers are part of the solutions to (94'), then the remaining variables of those solutions must satisfy the system

$$\begin{cases} \sum_{r=n+1}^k (w_r^l - z_r^l) = \lambda_l, & l = 1, \dots, n, \\ 0 \leq w_r \leq p^{-1}(P - u), \ 0 \leq z_r \leq p^{-1}(P - u), & r = n+1, \dots, k. \end{cases} \quad (98)$$

From properties (c) and (g), the number of solutions to (98) is $\leq J_{k-n}^{(n)}(\lfloor P/p \rfloor + 1)$. Combining the above discussions, we find that the number of solutions to (94'), i.e. the value of the multiple integral on the right-hand side of (93) is

$$\leq n!m^n p^{\frac{1}{2}n(n-1)} P^n J_{k-n}^{(n)} \left(\left\lfloor \frac{P}{p} \right\rfloor + 1 \right).$$

From this and (93) we obtain (89). □

Next, we estimate $J_{k,2}^{(n)}(P, p)$.

Lemma 7.3. *Let $p > n \geq 2$ be a prime, $k \geq n$. Then we have*

$$J_{k,2}^{(n)}(P, p) < \frac{2n^k}{(n-1)!} p^{k+n-1} J_k^{(n)} \left(\left\lfloor \frac{P}{p} \right\rfloor + 1 \right). \quad (99)$$

Proof. Let $\sum^*$ denote the summation over the variables $u_1, \dots, u_k, v_1, \dots, v_k$ satisfying: $1 \leq u_r \leq p, 1 \leq v_r \leq p, r = 1, \dots, k$, and among $u_1, \dots, u_k$ there are at most $n-1$ distinct values. Then, from the definition of $J_{k,2}^{(n)}(P, p)$ determined by (83), and by the symmetry of $u_1, \dots, u_k$ and $v_1, \dots, v_k$, it is easy to see that

$$J_{k,2}^{(n)}(P, p) \leq 2 \int_0^1 \dots \int_0^1 \sum^* S(u_1) \dots S(u_k) \bar{S}(v_1) \dots \bar{S}(v_k) d\alpha_1 \dots d\alpha_n,$$

where $S(u)$ is given by (92). Using Hölder's inequality we obtain

$$\begin{aligned} J_{k,2}^{(n)}(P, p) &\leq 2 \sum^* \left(\int_0^1 \dots \int_0^1 |S(u_1)|^{2k} d\alpha_1 \dots d\alpha_n \right)^{\frac{1}{2k}} \dots \left(\int_0^1 \dots \int_0^1 |S(v_k)|^{2k} d\alpha_1 \dots d\alpha_n \right)^{\frac{1}{2k}} \\ &\leq 2 \left(\sum^* 1 \right) \max_{1 \leq u \leq p} \int_0^1 \dots \int_0^1 |S(u)|^{2k} d\alpha_1 \dots d\alpha_n. \end{aligned} \quad (100)$$

It is easy to prove that

$$\sum^* 1 < C_p^{n-1} (n-1)^k p^k < \frac{n^k}{(n-1)!} p^{k+n-1}. \quad (101)$$

The last integral in (100) is equal to the number of solutions to the following system of congruences:

$$\begin{cases} \sum_{r=1}^k ((u + pw_r)^l - (u + pz_r)^l) = 0, & l = 1, \dots, n, \\ 1 \leq u + pw_r \leq P, \ 1 \leq u + pz_r \leq P, & r = 1, \dots, k, \end{cases} \quad (102)$$

where $u(1 \leq u \leq p)$ is a parameter. Clearly, (102) can be transformed into

$$\begin{cases} \sum_{r=1}^k (w_r^l - z_r^l) = 0, & l = 1, \dots, n, \\ 0 \leq w_r \leq p^{-1}(P - u), \quad 0 \leq z_r \leq p^{-1}(P - u), & r = 1, \dots, k. \end{cases} \quad (103)$$

The number of its solutions does not exceed $J_k^{(n)}(\lfloor P/p \rfloor + 1)$. From this, (101) and (100) the desired result is proved. $\square$

With Lemmas 7.2 and 7.3, we can now prove the main result of this section, which is usually called the Vinogradov-Hua Mean Value Theorem.

Theorem 7.4. *Let integers $n \geq 2, \tau \geq 0, k \geq n^2 + n\tau, P \geq 1$. Then we have $J_k^{(n)}(P) \leq (3n)^{2k\tau} P^{2k - \frac{1}{2}(n^2+n) + \delta_\tau}$, where $\delta = \delta_\tau = \frac{1}{2}(n^2 + n) \left(1 - \frac{1}{n}\right)^\tau$.*

Proof. We prove this by induction on τ . First assume $P \geq (2n)^{2n(1-\frac{1}{n})^{-\tau}}$. When $\tau = 0$, the inequality clearly holds. Suppose it holds for $\tau = d \geq 0$. We now prove it also holds for $\tau = d + 1$. $\square$

Since $P \geq (2n)^{2n}$, there must exist a prime p satisfying $2n^2 \leq \frac{1}{2}P^{\frac{1}{n}} \leq p \leq P^{\frac{1}{n}}$. Let $P_1 = \lfloor p^{-1}P \rfloor + 1$. Clearly, we have $P^{1-\frac{1}{n}} < P_1 < 3P^{1-\frac{1}{n}}$. Applying Lemma 7.2, since $m \leq \lfloor p^{-n}P \rfloor + 1 < 2p^{-n}P$, we obtain

$$J_{k,1}^{(n)}(P, p) \leq \frac{2^n}{n!} k^{2n} p^{2(k-n) - \frac{1}{2}(n^2+n)} P^{2n} J_{k-n}^{(n)}(P_1). \quad (104)$$

It is easy to verify that when n, k, P satisfy the conditions for $\tau = d + 1$, then $n, k - n, P_1$ satisfy the conditions for $\tau = d$. Thus, by the induction hypothesis, we have

$$J_{k-n}^{(n)}(P_1) \leq (3n)^{2(k-n)d} P_1^{2(k-n) - \frac{1}{2}(n^2+n) + \delta_d}. \quad (105)$$

Furthermore, using $P_1 < 3P^{1-\frac{1}{n}}$ and $p \leq P^{\frac{1}{n}}$, from the above two formulas, we derive

$$J_{k,1}^{(n)}(P, p) \leq \frac{2^n}{n!} k^{2n} (3n)^{2(k-n)d} 3^{2(k-n)} P^{2k - \frac{1}{2}(n^2+n) + \delta_{d+1}}. \quad (106)$$

Using $2 \leq n \leq p^{\frac{1}{2}}, k \geq n^2 + n$ (since $d + 1 \geq 1$), and $J_k^{(n)}(P_1) \leq P_1^{2n} J_{k-n}^{(n)}(P_1)$, from Lemma 7.3, we get

$$\begin{aligned} J_{k,2}^{(n)}(P, p) &< \frac{2n^{2n}}{(n-1)!} n^{k-2n} p^{k+n-1} J_k^{(n)}(P_1) \\ &< \frac{n^{2n}}{n!} p^{\frac{1}{2}k} P_1^{2n} J_{k-n}^{(n)}(P_1) \\ &< \frac{n^{2n}}{n!} P^{2k - \frac{1}{2}(n^2+n)} P^{2n} J_{k-n}^{(n)}(P_1). \end{aligned}$$

Furthermore, from this and (105) we obtain

$$J_{k,2}^{(n)}(P, p) < \frac{n^{2n}}{n!} 3^{2k} (3n)^{2(k-n)d} P^{2k - \frac{1}{2}(n^2+n) + \delta_{d+1}}. \quad (107)$$

Thus, from (83), (106) and (107), we obtain (noting that $k > n \geq 2$):

$$J_k^{(n)}(P) < 3^{2k} k^{2n} (3n)^{2(k-n)d} P^{2k - \frac{1}{2}(n^2+n) + \delta_{d+1}} < (3n)^{2k(d+1)} P^{2k - \frac{1}{2}(n^2+n) + \delta_{d+1}}, \quad (108)$$

This proves that the inequality also holds for $\tau = d + 1$. Therefore, the theorem holds under the condition $P \geq (2n)^{2n(1-\frac{1}{n})^{-\tau}}$.

Next, we prove that the theorem also holds when $P < (2n)^{2n(1-\frac{1}{n})^{-\tau}}$. We still use induction on τ . When $\tau = 0$, the inequality clearly holds. Suppose it holds for $\tau = d \geq 0$. We now prove that it also holds for $\tau = d + 1$. We discuss two cases: (a) $P \geq (2n)^{2n(1-\frac{1}{n})^{-d}}$, and (b) $P < (2n)^{2n(1-\frac{1}{n})^{-d}}$.

When case (a) holds, using the previously proved conclusion to estimate $J_{k-n}^{(n)}(P)$ (the case $\tau = d$), from the inequality $J_k^{(n)}(P) \leq P^{2n} J_{k-n}^{(n)}(P)$, we obtain

$$J_k^{(n)}(P) \leq P^{2n} (3n)^{2(k-n)d} P^{2(k-n) - \frac{1}{2}(n^2+n) + \delta_d} < (3n)^{2k(d+1)} P^{2k - \frac{1}{2}(n^2+n) + \delta_{d+1}}. \quad (109)$$

This proves that the inequality also holds for $\tau = d + 1$. In the last step, we used $k \geq n^2 + n$ (since $d + 1 \geq 1$) and

$$P^{\delta_d - \delta_{d+1}} = P^{\frac{1}{2}(n+1)(1-\frac{1}{n})^d} < \left((2n)^{2n(1-\frac{1}{n})^{-d-1}} \right)^{\frac{1}{2}(n+1)(1-\frac{1}{n})^d} \leq (2n)^{2k}.$$

When case (b) holds, by using the induction hypothesis to estimate $J_{k-n}^{(n)}(P)$, we likewise obtain (109). Therefore, the inequality also holds for $\tau = d + 1$. Thus, the theorem also holds when $P < (2n)^{2n(1-\frac{1}{n})^{-\tau}}$. The proof is complete.

8 Estimates of Weyl Sums

Lemma 8.1. *Let k, Y be arbitrary positive integers, $Y < P$. Furthermore, let*

$$\alpha_{n+1} = \frac{a}{q} + \frac{\theta}{q^2}, \quad q > 1, \quad (a, q) = 1, \quad |\theta| \leq 1. \quad (110)$$

Then we have

$$S(\alpha_{n+1}) \ll \left\{ n(2k)^n P^{2k + \frac{n(n+1)}{2}} \left(\frac{1}{q} + \frac{1}{Y} + \frac{q \log q}{Y P^n} \right) \cdot J_k^{(n)}(P) \right\}^{\frac{1}{2k}} + Y, \quad (111)$$

where the constant in $\ll$ is an absolute constant.

Proof. Clearly, we have

$$S(\alpha_{n+1}) = Y^{-1} \sum_{y=1}^Y \sum_{x=1}^{P-y} e(f(x+y)) = W + O(Y), \quad (112)$$

where

$$W = Y^{-1} \sum_{y=1}^Y \sum_{x=1}^P e(f(x+y)). \quad (113)$$

Writing $f(x+y)$ as

$$f(x+y) = \alpha_{n+1}x^{n+1} + g_n(y)x^n + \cdots + g_1(y)x + g_0(y), \quad (114)$$

where $g_n(y) = (n+1)\alpha_{n+1}y + \alpha_n$. Thus, using Hölder's inequality, we get

$$\begin{aligned} |W|^{2k} &\leq Y^{-1} \sum_{y=1}^Y \left| \sum_{x=1}^P e(\alpha_{n+1}x^{n+1} + g_n(y)x^n + \cdots + g_1(y)x) \right|^{2k} \\ &= Y^{-1} \sum_{y=1}^Y \sum_{\lambda_1, \dots, \lambda_{n+1}} J_k^{(n+1)}(P; \lambda_1, \dots, \lambda_{n+1}) \cdot e(\alpha_{n+1}\lambda_{n+1} + g_n(y)\lambda_n + \cdots + g_1(y)\lambda_1), \end{aligned}$$

where the second summation is over all possible values of $\lambda_1, \dots, \lambda_{n+1}$. From this and

$$\sum_{\lambda_{n+1}} J_k^{(n+1)}(P; \lambda_1, \dots, \lambda_{n+1}) = J_k^{(n)}(P; \lambda_1, \dots, \lambda_n), \quad (115)$$

and applying the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned} |W|^{2k} &\leq Y^{-1} \sum_{\lambda_1, \dots, \lambda_n} J_k^{(n)}(P; \lambda_1, \dots, \lambda_n) \cdot \left| \sum_{y=1}^Y e(g_n(y)\lambda_n + \cdots + g_1(y)\lambda_1) \right| \\ &\leq Y^{-1} \left(\sum_{\lambda_1, \dots, \lambda_n} (J_k^{(n)}(P; \lambda_1, \dots, \lambda_n))^2 \right)^{\frac{1}{2}} \cdot \left(\sum_{\lambda_1, \dots, \lambda_n} \left| \sum_{y=1}^Y e(g_n(y)\lambda_n + \cdots + g_1(y)\lambda_1) \right|^2 \right)^{\frac{1}{2}} \\ &\leq Y^{-1} P^k (J_k^{(n)}(P))^{\frac{1}{2}} \cdot \left(\sum_{\lambda_1, \dots, \lambda_n} \left| \sum_{y=1}^Y e(g_n(y)\lambda_n + \cdots + g_1(y)\lambda_1) \right|^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (116)$$

The last step utilized properties (c) and (e) of Section 7.

Next, we use the properties of Section 7 and the estimates for linear exponential sums to estimate the last sum in (116). From property (d) of Section 7, we know $|\lambda_l| < kP^l$, $l = 1, \dots, n$. Using (110) we get

$$\begin{aligned} &\sum_{\lambda_1, \dots, \lambda_n} \left| \sum_{y=1}^Y e(g_n(y)\lambda_n + \cdots + g_1(y)\lambda_1) \right|^2 \\ &= \sum_{y=1}^Y \sum_{y'=1}^Y \sum_{\lambda_1, \dots, \lambda_n} e((g_n(y) - g_n(y'))\lambda_n + \cdots + (g_1(y) - g_1(y'))\lambda_1) \\ &\leq \sum_{y=1}^Y \sum_{y'=1}^Y \left| \sum_{\lambda_n} e((n+1)\alpha_{n+1}(y - y')\lambda_n) \right| \sum_{\lambda_1, \dots, \lambda_{n-1}} 1 \\ &\leq (2k)^{n-1} P^{\frac{1}{2}n(n-1)} \cdot \sum_{y=1}^Y \sum_{y'=1}^Y \min \left(2kp^n, \frac{1}{\|(n+1)\alpha_{n+1}(y - y')\|} \right) \end{aligned}$$

$$\begin{aligned}
&\leq (2k)^{n-1} P^{\frac{1}{2}n(n-1)} \cdot \sum_{y=1}^Y \sum_{y'=1}^Y \min \left(2kp^n, \frac{1}{\|\alpha_{n+1}y' - (n+1)\alpha_{n+1}y\|} \right) \\
&\ll (2k)^{n-1} P^{\frac{1}{2}n(n-1)} \sum_{y=1}^Y \left(\frac{(n+1)Y}{q} + 1 \right) \cdot (2kp^n + q \log q) \\
&\ll n(2k)^n P^{\frac{1}{2}n(n+1)} Y^2 \left(\frac{1}{q} + \frac{1}{Y} + \frac{q \log q}{Y P^n} + \frac{\log q}{P^n} \right) \\
&\ll n(2k)^n P^{\frac{1}{2}n(n+1)} Y^2 \left(\frac{1}{q} + \frac{1}{Y} + \frac{q \log q}{Y P^n} \right). \tag{117}
\end{aligned}$$

In the last step, we used the fact that when $q < Y$, we have $\frac{\log q}{P^n} < \frac{\log Y}{Y^2} \ll \frac{1}{Y}$.

From (112), (116), and (117), we immediately obtain (111). $\square$

From Lemma 8.1 and Theorem 7.4, we immediately get the following theorem.

Theorem 8.2. *Let integers $\tau \geq 0, k \geq n^2 + n\tau$. Then, under the conditions of Lemma 1, we have*

$$S(\alpha_{n+1}) \ll (3n)^{\frac{1}{2}} P \left\{ P^{\delta_\tau} \left(\frac{1}{q} + \frac{1}{Y} + \frac{q \log q}{Y P^n} \right) \right\}^{\frac{1}{4k}} + Y, \tag{118}$$

where the constant in $\ll$ is an absolute constant, and δ_τ is given by Theorem 7.4.

Proof. Substituting estimate into (111), and noting that $(n(2k)^n)^{\frac{1}{2k}} \ll 1$. $\square$

Theorem 8.3. *Let $n \geq 2$, and let α_{n+1} be given by (3). Then when $P^{\frac{1}{n}} \leq q \leq P^{n+1-\frac{1}{4}}$, we have*

$$S(\alpha_{n+1}) \ll (3n)^{3n \log n} P^{1-(400n^2 \log n)^{-1}}, \tag{119}$$

where the constant in $\ll$ is an absolute constant.

Proof. Take $\tau = \lfloor 4n \log n \rfloor + 1$, $k = n^2 + n\tau$, and $Y = \lfloor P^{1-1/(4n^2)} \rfloor$. At this time, we have

$$\begin{aligned}
\delta_\tau &= \frac{1}{2}(n^2 + n) \left(1 - \frac{1}{n} \right)^\tau < \frac{1}{2}(n^2 + n) \left(1 - \frac{1}{n} \right)^{4n \log n} \leq \frac{1}{2}(n^2 + n)n^{-4} \leq \frac{3}{16}, \quad n \geq 2. \\
\frac{1}{q} + \frac{1}{Y} + \frac{q \log q}{Y P^n} &\ll n P^{-\frac{1}{4}+1/(40n^2)} \log P \ll n P^{-\frac{19}{80}}, \quad n \geq 2.
\end{aligned}$$

Noting that $k \leq 5n^2 \log n$, substituting the above two formulas and the chosen values into (118), we obtain the desired result. $\square$

From the above proofs, we can see the basic idea of Vinogradov's method for estimating Weyl sums: First, using some method, the Weyl sum to be estimated is reduced to estimating a double exponential sum $\sum_x \sum_y e(F(x, y))$,

where $F(x, y)$ is a bivariate polynomial in x and y ; Then using Hölder's inequality, the estimate of the exponential sum is transformed into estimating the mean value $J_k^{(n)}(P)$ and some linear exponential sum; Finally using Vinogradov's mean value theorem and his method for estimating linear exponential sums, the desired estimate of the Weyl sum is obtained. This method can take different forms in applications. In the next part, we will use this method to estimate ζ -sum $\sum n^{it}$. Formally, the proof in the next section differs somewhat from this section, but the method is the same.

Let $1 \leq N < N' \leq 2N$. We estimate the Weyl sum

$$S(N) = \sum_{N < n \leq N'} n^{it} = \sum_{N < n \leq N'} e\left(\frac{t \log n}{2\pi}\right). \tag{120}$$

It is also usually called the ζ -sum. Below, we will prove

Theorem 8.4. *There exist absolute positive constants c_1, c_2 such that when $1 \leq N \leq t$, we have*

$$|S(N)| \leq c_1 N \exp\left(-c_2 \frac{\log^3 N}{\log^2 t}\right). \tag{121}$$

First, we prove a few lemmas.

Lemma 8.5. Let $100 \leq N \leq t, 0 < d < 1$. Furthermore, let $a = \left\lfloor N^{\frac{d}{2}} \right\rfloor$, $r = \left\lceil \frac{\log t}{(1-d)\log N} \right\rceil + 1$. Then we have

$$|S(N)| \leq a^{-2} \sum_{n \leq n \leq N'} |W_1(n, a)| + O(N^d), \quad (122)$$

where

$$W_1(n, a) = \sum_{x=1}^a \sum_{y=1}^a e(\alpha_r x^r y^r + \cdots + \alpha_1 xy), \quad \alpha_m = \frac{(-1)^{m-1} t}{2\pi m n^m}, \quad m = 1, 2, \dots, r.$$

Proof. For any integers x, y , we have

$$S(N) = \sum_{N-xy < n \leq N'-xy} e\left(\frac{t \log(n+xy)}{2\pi}\right) = \sum_{N < n \leq N'} e\left(\frac{t \log(n+xy)}{2\pi}\right) + O(|xy|).$$

Furthermore, we have

$$a^2 S(N) = \sum_{x=1}^a \sum_{y=1}^a \sum_{N < n \leq N'} e\left(\frac{t \log(n+xy)}{2\pi}\right) + O(a^4).$$

Hence,

$$|S(N)| \leq a^{-2} \sum_{N < n \leq N'} \left| \sum_{x=1}^a \sum_{y=1}^a e\left(\frac{t}{2\pi} \log\left(1 + \frac{xy}{n}\right)\right) \right| + O(a^2) = a^{-2} \sum_{N < n \leq N'} |W(n, a)| + O(N^d), \quad (123)$$

where

$$W(n, a) = \sum_{x=1}^a \sum_{y=1}^a e\left(\frac{t}{2\pi} \log\left(1 + \frac{xy}{n}\right)\right). \quad (124)$$

Since when $1 \leq x, y \leq a = \lfloor N^{d/2} \rfloor, n > N$, and r is given by (111), we have $\frac{xy}{n} < a^2 N^{-1} \leq N^{d-1} < 1$, and

$$t \left(\frac{xy}{n}\right)^{r+1} < t N^{-(1-d)(r+1)} = \exp\left\{-(1-d) \cdot \left(r+1 - \frac{\log t}{(1-d)\log N}\right) \log N\right\} < N^{d-1}.$$

Thus, by Taylor's formula, we obtain

$$t \log\left(1 + \frac{xy}{n}\right) = t F_r\left(\frac{xy}{n}\right) + O(N^{d-1}), \quad (125)$$

$$e\left(\frac{t}{2\pi} \log\left(1 + \frac{xy}{n}\right)\right) = e\left(\frac{t}{2\pi} F_r\left(\frac{xy}{n}\right)\right) + O(N^{d-1}). \quad (126)$$

where $F_r(z) = z - \frac{1}{2}z^2 + \cdots + \frac{(-1)^{r-1}}{r}z^r$. Thus, from (115) we get $W(n, a) = W_1(n, a) + O(a^2 N^{d-1})$. From this the desired result follows. $\square$

Next we estimate the bivariate polynomial exponential sum $W_1(n, a)$.

Lemma 8.6. Under the conditions and notation of Lemma 8.2, for a positive integer k , we have

$$|W_1(n, a)|^{4k^2} \leq a^{8k^2-4k} (J_k^{(r)}(a))^2 \prod_{m=1}^r \sum_{|\mu_m| < k a^m} \left| \sum_{|\lambda_m| < k a^m} e(\alpha_m \mu_m \lambda_m) \right|. \quad (127)$$

Proof. Applying Hölder's inequality twice successively to (113), we obtain

$$\begin{aligned} |W_1(n, a)|^{4k^2} &\leq a^{4k^2-2k} \left\{ \sum_{\lambda_1, \dots, \lambda_r} \left(J_k^{(r)}(a; \lambda_1, \dots, \lambda_r) \right)^{\frac{2k}{2k-1}} \right\}^{2k-1} \cdot \sum_{\substack{\lambda_1, \dots, \lambda_r \\ |\lambda_m| < k a^m}} \left| \sum_{x=1}^a e(\alpha_r \lambda_r x^r + \cdots + \alpha_1 \lambda_1 x) \right|^{2k} \\ &\leq a^{8k^2-4k} J_k^{(r)}(a) \cdot \sum_{\substack{\lambda_1, \dots, \lambda_r \\ |\lambda_m| < k a^m}} \left| \sum_{x=1}^a e(\alpha_r \lambda_r x^r + \cdots + \alpha_1 \lambda_1 x) \right|^{2k}. \end{aligned} \quad (128)$$

Here we used properties (c), (d), and (e) of Section 7. Furthermore, using these properties, we obtain

$$|W_1(n, a)|^{4k^2} = a^{8k^2-4k} J_k^{(r)}(a) \sum_{\substack{\lambda_1, \dots, \lambda_r \\ |\lambda_m| < k a^m}} \sum_{\substack{\mu_1, \dots, \mu_r \\ |\mu_m| < k a^m}} J_k^{(r)}(a; \mu_1, \dots, \mu_r) e(\alpha_r \mu_r \lambda_r + \cdots + \alpha_1 \mu_1 \lambda_1)$$

$$\leq a^{8k^2-4k} (J_k^{(r)}(a))^2 \sum_{\substack{\mu_1, \dots, \mu_r \\ |\mu_m| < ka^m}} \left| \sum_{\substack{\lambda_1, \dots, \lambda_r \\ |\lambda_m| < ka^m}} e(\alpha_r \mu_r \lambda_r + \dots + \alpha_1 \mu_1 \lambda_1) \right|. \quad (129)$$

This proves (127). $\square$

Thus, the estimate of $W_1(n, a)$ is transformed into estimating the product of linear exponential sums on the right-hand side of (127).

Lemma 8.7. *Under the conditions and notation of Lemmas 8.5 and 8.6, if we take $d = \frac{10}{11}$, then when $k^2 \geq r$, we have*

$$\prod_{m=1}^r \sum_{|\mu_m| < ka^m} \left| \sum_{|\lambda_m| < ka^m} e(\alpha_m \mu_m \lambda_m) \right| \leq c'_3 8^{r^2} (2k)^{2r} a^{r^2+r} \exp\left(-\frac{4 \log^2 t}{5 \log N}\right), \quad (130)$$

where c_3 is an absolute positive constant.

Proof.

$$\left| \sum_{|\lambda_m| < ka^m} e(\alpha_m \mu_m \lambda_m) \right| \leq \min\left(2ka^m, \frac{2}{\|\alpha_m \mu_m\|}\right).$$

If α_m can be expressed as

$$\alpha_m = \frac{a_m}{q_m} + \frac{\theta_m}{q_m^2}, \quad q_m \geq 3, \quad (q_m, a_m) = 1, \quad |\theta_m| \leq 1, \quad (131)$$

then for $1 \leq m \leq r$, we have

$$\sum_{|\mu_m| < ka^m} \left| \sum_{|\lambda_m| < ka^m} e(\alpha_m \mu_m \lambda_m) \right| \ll (2ka^m)^2 \min\left(1, \left(\frac{1}{q_m} + \frac{1}{2ka^m} + \frac{q_m}{(2ka^m)^2}\right) \log q_m\right). \quad (132)$$

In order for the left-hand side of the above formula to obtain an estimate better than the trivial estimate $(2ka^m)^2$, q_m must be taken as large as possible, but smaller than the order of $(2ka^m)^2$. Therefore, we need to further discuss the representation (131) for these α_m .

Let $m = rh$.

$$t^{\frac{1}{1-d}} < N^m = N^{rh} < t^{h(\frac{1}{1-d}+1)}, \quad 1 \leq m \leq r. \quad (133)$$

Therefore, when $1 \leq m \leq r$,

$$2\pi m t^{\frac{h}{1-d}-1} < \frac{1}{|\alpha_m|} < 2^{m+1} \pi m t^{h(\frac{1}{1-d}+1)}. \quad (134)$$

Therefore, when $h \geq 1-d$, i.e., $m \geq (1-d)r$, we have $|\alpha_m| < \frac{1}{6}$. Now take $q_m = \lfloor |\alpha_m|^{-1} \rfloor$, $a_m = (-1)^{m-1}$. Then when $(1-d)r \leq m \leq r$,

$$\alpha_m = \frac{(-1)^{m-1}}{q_m} + \frac{\theta_m}{q_m^2}, \quad q_m \geq 6, \quad |\theta_m| \leq 1. \quad (135)$$

From (133) and (134), we know that when $(1-d)r \leq m \leq r$,

$$\begin{cases} q_m > \frac{1}{2|\alpha_m|} > \pi m t^{\frac{h}{1-d}-1}, \\ 2ka^m > 2^{1-m} k N^{\frac{md}{2}} > 2^{1-m} k t^{\frac{hd}{2(1-d)}}, \\ \frac{(2ka^m)^2}{q_m} > \frac{2k^2}{\pi m 8^m} N^{-(1-d)m} t > \frac{2k^2}{\pi m 8^m} t^{1-k(2-d)}. \end{cases} \quad (136)$$

From this, we can see that in order for (132) to yield a non-trivial estimate (ignoring the coefficient first), we only need $1-d < h < (2-d)^{-1}$, i.e., $(1-d)r < m < (2-d)^{-1}r$. Clearly, as long as d is suitably close to 1, this range of values for m is non-empty. Now take $d = \frac{10}{11}$. Then when $k^2 \geq r$,

$$\frac{2}{11}r = 2(1-d)r \leq m \leq \frac{1}{2}(2-d)^{-1}r = \frac{11}{24}r. \quad (137)$$

From (132) and (136), we obtain

$$\sum_{|\mu_m| < ka^m} \left| \sum_{|\lambda_m| < ka^m} e(\alpha_m \mu_m \lambda_m) \right| \ll 8^m (2ka^m)^2 t^{-\frac{1}{2}} \log q_m \ll 8^m (2ka^m)^2 t^{-\frac{1}{2}} \log t \leq c_4 8^m (2ka^m)^2 t^{-\frac{2}{3}}, \quad (138)$$

where c_4 is a positive constant, and we used $\log q_m \leq \log \frac{1}{|\alpha_m|} \ll \log t$, $1 \leq m \leq r$. From (132) and (138), it follows that when $d = \frac{10}{11}, k^2 > r$, the left-hand side of (130) is

$$\leq \prod_{m=1}^r c_5 (2ka^m)^2 \prod_{\frac{2}{11}r \leq m \leq \frac{11}{24}r} 8^m t^{-\frac{2}{3}} \leq c'_5 8^{r^2} (2k)^{2r} a^{r^2+r} \exp\left(-\frac{4}{5} \frac{\log^2 t}{\log N}\right).$$

This proves the desired conclusion. $\square$

Proof of Theorem 8.4. Without loss of generality, assume $100 \leq N$. Under the conditions and notation of Lemma 8.5, take $d = \frac{10}{11}, \tau = 6r, k = r^2 + r\tau = 7r^2$. From (127), (130), and Theorem 7.4, we get

$$|W_1(n, a)|^{4k^2} \leq c'_5 8^{r^2} (3r)^{4kr} (2k)^{2r} a^{8k^2} a^{2\delta_\tau} \cdot \exp\left(-\frac{4}{5} \frac{\log^2 t}{\log N}\right),$$

where $2\delta_\tau = (r^2 + r) \left(1 - \frac{1}{r}\right)^\tau$. From $a \leq N^{\frac{5}{11}}, r \leq 12 \frac{\log t}{\log N}$, we have

$$a^{2\delta_\tau} \leq \exp\left(71 \left(1 - \frac{1}{r}\right)^\tau \frac{\log^2 t}{\log N}\right) \leq \exp\left(\frac{1}{5} \frac{\log^2 t}{\log N}\right),$$

Here we used $\tau = 6r$ and $71 \left(1 - \frac{1}{r}\right)^\tau = 71e^{-6} < \frac{1}{5}$. From the above two formulas, we get

$$|W_1(n, a)| \leq c_6 a^2 \exp\left(-\frac{3}{20k^2} \frac{\log^2 t}{\log N}\right) \leq c_6 a^2 \exp\left(-\frac{3}{980 \cdot 12^4} \frac{\log^3 N}{\log^2 t}\right). \quad (139)$$

From this and (122), Theorem 8.4 is proved. $\square$

Corollary 8.8. *Let $1 \leq N \leq t$. Then there exist absolute positive constants c_7, c_8 such that*

$$\left| \sum_{n=1}^N n^{it} \right| \leq c_7 N \exp\left(-c_8 \frac{\log^3 N}{\log^2 t}\right). \quad (140)$$

Proof. Take an integer k satisfying $2^{-(k+1)}N < \sqrt{N} \leq 2^{-k}N$. From Theorem 8.4, we get

$$\sum_{n=1}^N n^{it} = \sum_{n \leq 2^{-k}N} n^{it} + \sum_{j=0}^{k-1} \sum_{-(j+1)N < n \leq 2^{-j}N} n^{it} \ll \sqrt{N} + \sum_{j=0}^{k-1} \frac{N}{2^{j+1}} \exp\left(-c_2 \frac{\log^3 \sqrt{N}}{\log^2 t}\right),$$

From this, the desired conclusion follows. $\square$

Completely identical to the proofs of Theorem 8.4 and Corollary 8.8, we can prove

Theorem 8.9. *Let $1 \leq N \leq t, 0 \leq \theta < 1$. There exist absolute constants $c_9, c_{10}, c_{11}, c_{12}$ such that*

$$\left| \sum_{N < n \leq N'} (n + \theta)^{it} \right| \leq c_9 N \exp\left(-c_{10} \frac{\log^3 N}{\log^2 t}\right), \quad N < N' \leq 2N, \quad (141)$$

and

$$\left| \sum_{n=1}^N (n + \theta)^{it} \right| \leq c_{11} N \exp\left(-c_{12} \frac{\log^3 N}{\log^2 t}\right). \quad (142)$$

To prove Theorem 8.9, it suffices to replace n (excluding the summation index) in the proofs of Theorem 8.4 and Corollary 8.8 with $n + \theta$; everything else is the same.